New IND-CPA-D Attacks on BFV and BGV*

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Abstract

IND-CPA-D security is an extension of IND-CPA security introduced in the context of Fully Homomorphic Encryption (FHE). It captures the situation where the computation result is known to the adversary; it is in particular necessary for FHE with threshold decryption. This work presents three new attacks in this situation on the BFV and BGV FHE schemes, when their parameters are instantiated with heuristic assumptions commonly used to enable correctness.

We first extend the *bootstrapping failure attacks* from Cheon *et al.* [CCS'24] on the DM/CGGI and CKKS schemes, to both BFV and BGV. We focus on BGV when instantiated with the so-called powerful-basis representation of ring elements. The second attack relies on *skewing the ciphertext noise* in BFV, making it outpace the heuristic noise bounds for plaintext-ciphertext multiplications. This is achieved by crafting plaintexts, for which the noise bounds do not hold. The third and last attack focuses on outpacing the noise growth in BFV ciphertext-ciphertext multiplications. For this purpose, we *skew the mask* part of the ciphertext by selecting a random encryption of zero, whose addition yields a ciphertext with a sufficiently large mask norm. While the first two attacks successfully recover the secret key, the last attack instead breaks the IND-CPA-D indistinguishability game.

We report experiments on all these attacks, based on the HElib and OpenFHE libraries, respectively, for BGV and BFV. Notably, the second attack implemented in OpenFHE fully recovers the secret key in a few minutes to a few hours, depending on the parameters.

CCS Concepts

• **Security and privacy → Cryptanalysis and other attacks**.

Keywords

Fully Homomorphic Encryption, BGV, BFV, IND-CPA-D Security

1 Introduction

IND-CPA-D security was introduced by Li and Micciancio [LM21] to model an IND-CPA attacker against a fully homomorphic encryption (FHE) scheme, who would have access to a decryption oracle for honestly created ciphertexts. While Li and Micciancio focused on the CKKS scheme for approximate computations on real/complex numbers [CKKS17], it was subsequently shown that IND-CPA-secure FHE schemes for exact arithmetic (BGV [BGV12], BFV [Bra12, FV12] and DM/CGGI [DM15, CGGI16]) may be IND-CPA-D insecure when perfect correctness is not achieved [CSBB24, CCP+24]. These works also observed that IND-CPA-D attacks directly impact the security of (both exact and approximate) Threshold-FHE schemes [AJLA+12, BGG+18] with imperfect correctness.

Perfect correctness requires that the decryption of honestly created ciphertexts (through legitimate uses of the encryption and evaluation functions) produces the plaintext that would have been obtained by running the computation in clear, or a nearby plaintext in case of FHE for approximate computations, with probability 1 or extremely close to it over the randomness used by the FHE algorithms. Unfortunately, perfect correctness is complex and costly to achieve, and most FHE libraries (including OpenFHE [Ope25], HElib [HEl23], SEAL [CLP17], Lattigo [Lat24] and HEaaN [HEa24]) do not provide it. Some attacks [GNSJ24, CSBB24, CCP+24] have exploited lacks of proper checks on how many additions can be performed in a row. These can be prevented by enforcing such limitations, such as controlling how many additions can be performed without a multiplication, or by parameterizing the FHE scheme specifically for a given circuit or family of circuits [ABMP26]. Attacks described in [CCP+24] exploited the failures in the DM/CGGI and CKKS bootstrapping algorithms: bootstrapping is a ciphertext maintenance procedure that is so costly that it is very tempting to weaken correctness to increase performance. Subsequently, modifications of DM/CGGI have been proposed in [BJSW25, dRDV26] to make bootstrapping failures much less frequent, while the same can be achieved for CKKS by using the sparse secret encapsulation technique from [BTPH22]. Finally, in [LWF25], the authors mounted an IND-CPA-D attack based on bootstrapping errors for out-of-range plaintexts in a variant of BFV/BGV bootstrapping [LW24].

Even when bootstrapping correctness is tightened and the shape of allowed circuits is controlled, all FHE implementations that we are aware of remain only *heuristically* correct due to how ciphertext noise is bounded. The main FHE schemes rely on (variants of) the LWE [Reg05] and RLWE problems [SSTX09, LPR10], where ciphertext computations are inherently noisy. To ensure correctness, one must bound the noise growth during elementary computations to set the scheme parameters. While the worst-case bound for the product of two elements $e, f \in \mathcal{R} = \mathbb{Z}[X]/(X^N + 1)$ for a power-of-two N is $\|e \cdot f\|_\infty \leq N \cdot \|e\|_\infty \cdot \|f\|_\infty$, most FHE libraries (e.g., [HEl23, CLP17, Ope25, Lat24]) adopt much smaller bounds to reduce parameters and improve performance.

The most prevalent heuristic assumes a "random-like" behavior of noise and ciphertext elements, effectively asserting that $\|e \cdot f\|_\infty \lesssim c\sqrt{N} \cdot \|e\|_\infty \cdot \|f\|_\infty$ for some small constant $c > 0$. In practice, this saving is often justified by reasoning that if coefficients were independently and uniformly sampled, one could obtain this $\sqrt{N}$-factor saving rigorously via Hoeffding-like inequalities.

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This heuristic is manifested in various forms: some works explicitly use a $\delta_N = \mathcal{O}(\sqrt{N})$ factor for L^∞-norm bounds (e.g., [KPZ21, Section 2], [HPS19, Section 6.1], and [GZ25, Section 3.3]), while others adopt variance-based approaches (e.g., [CS16, HS20, BMMU25, BCF+25, BMCM23]). These bounds implicitly rely on the critical assumption that the polynomials do not "align" in a worst-case manner in practice.

However, this critical assumption, that polynomials will not align in a worst-case manner, is increasingly untenable in the IND-CPA-D setting. While heuristic "savings" might hold for honest, randomly distributed inputs, they do not account for the potential for adversarial influence. Since homomorphic operations are entirely deterministic, the only randomness comes from the honest generation of the input ciphertexts. The adversary does not control their distribution, but it does choose which of them to combine and in what order; this alone suffices to steer the noise away from its average-case behavior, so that heuristic independence across the computation no longer holds.

The fundamental issue lies in the misapplication of IND-CPA correctness analysis to the IND-CPA-D setting. Standard error-estimation tools inherently assume that inputs, such as plaintexts or rounding errors, are benign and independent of the secret key. Early indications of this vulnerability have appeared in certain restricted and stylized settings. For example, prior work has demonstrated that violating independence assumptions, such as setting two independent variables to take the same value, can invalidate noise bounds and expose secret information, leading to key-recovery attacks under a decryption oracle (KR-D) [CCP+24, CSBB24], which directly imply IND-CPA-D insecurity. Furthermore, while recent literature has noted that correlations between noise terms can lead to heavy-tailed distributions [MP20, BMCM23, BMMU25, BCM+25, GZ25], the discrepancy between these heuristic bounds and actual noise behavior has not yet been fully exploited from a cryptanalytic perspective.

In this work, we demonstrate that these heuristic savings are not merely theoretical inaccuracies or statistical curiosities, but provide a concrete bridge for mounting practical IND-CPA-D and KR-D attacks. By proactively "twisting" the input distribution toward its worst-case expansion, an adversary can systematically defeat the noise-growth assumptions used in prior works and adopted by major FHE libraries.

1.1 Contributions

We extend the IND-CPA-D cryptanalysis landscape by exhibiting *three new attacks* on exact FHE schemes that exploit correctness imperfectness.

The first attack is a KR-D attack that exploits potential *bootstrapping failures* for BGV and BFV. Together with the bootstrapping failure attacks from [CCP+24], this shows that the bootstrapping algorithms of all major FHE schemes are susceptible to IND-CPA-D attacks. We then identify two critical vulnerabilities in current noise analysis paradigms that target the underlying assumption that plaintexts and ciphertexts behave as random, independent variables. This leads to our second and third attacks, which are the first to exploit the *heuristic nature of noise analysis*. While many implementations and analyses (e.g., [BCF+25, BMCM23, BMMU25, CS16]) rely on probabilistic savings or variance-based estimates to set parameters, we demonstrate that these estimates are fundamentally fragile against adversaries in the IND-CPA-D setting. By crafting inputs that bypass these statistical assumptions to maximize noise expansion, our second attack leverages these vulnerabilities in BFV plaintext-ciphertext (pt-ct) multiplication to mount a KR-D attack that fully and efficiently recovers secret keys. We then show how to extend this methodology to BFV ciphertext-ciphertext (ct-ct) multiplications, resulting in an efficient IND-CPA-D attack for the third case.

These three attacks are supported by various experimental results. The bootstrapping failure attack was validated on HElib [HEl23] using parameters specifically chosen to improve its efficacy, indicating that some of the original parameters offer only about 80 bits of IND-CPA-D security. The BFV pt-ct attack was successfully executed on OpenFHE [Ope26] without any modifications to the library, fully recovering the secret keys within several minutes to a few hours. For the third attack, the required fresh encryptions were simulated using the secret key; due to the substantial search complexity, this resulted in an IND-CPA-D attack rather than complete key recovery. Apart from this emulation step, the attack was performed on OpenFHE with its default configuration.

1.2 Technical Overview

Our first attack is inspired by the key-recovery attack of [CCP+24], which targets exact FHE schemes such as DM/CGGI and discrete versions of CKKS by exploiting bootstrapping failures. We extend this line of attacks to FHE schemes over finite fields by analyzing the correctness conditions of BGV/BFV bootstrapping and showing how approximate inequality hints lead to secret key leakage. More concretely, BGV/BFV bootstrapping typically relies on digit extraction preceded by a modulus switching and linear transformations. Similar to the vulnerabilities observed in [CCP+24], if the error incurred during modulus switching exceeds a specific threshold, it compromises the digit extraction process, leading to a bootstrapping failure. However, the non-power-of-two cyclotomic rings and the *powerful basis* multivariate polynomial structure used in HElib make the translation of these errors into inequality hints more involved: each powerful-basis coordinate of the error is still a fixed linear function of the secret key, but a single secret coordinate now influences a failure through several such coordinates rather than one. By uncovering the underlying algebraic structure, we observe that the failure is instead induced by Irwin-Hall-like variables derived from the modulus switching noise. We thus model the failure condition through these variables to recover the non-zero indices of the secret key. As the secret key is sparse, it can be efficiently recovered from the non-zero indices with LWE solvers in small dimension. This provides a key-recovery attack in the IND-CPA-D model, against BGV and its bootstrapping implementation in the HElib library.

The subsequent attacks are built upon our identification of two critical vulnerabilities in current noise analysis paradigms, where heuristic bounds are fundamentally fragile in the IND-CPA-D setting. Even within the *regime of legitimate evaluation*, an adversary can maliciously craft inputs to exploit the gap between heuristic expectations and worst-case behavior.

Our second attack addresses the noise estimation of plaintext-ciphertext (pt-ct) multiplications in BFV by invalidating the heuristic saving $\|\varepsilon_{\mathrm{pt-ct}}\|_\infty = \|\mu \cdot \varepsilon\|_\infty \lesssim \delta_N \|\mu\|_\infty \cdot \|\varepsilon\|_\infty$ where $\delta_N = \mathcal{O}(\sqrt{N})$. While standard IND-CPA analyses assume that plaintexts μ are sampled from a distribution behaving like random noise (e.g., [BMMU25, page 2]), we demonstrate that an adversary can select a specific μ to maximize the expansion factor. This allows the adversary to force the noise to exceed heuristic bounds, such as, with canonical norm, $\|\varepsilon_{\mathrm{pt-ct}}\|_\infty^{\mathrm{can}} \leq 6\sqrt{N(t^2-1)/12} \cdot \|\varepsilon\|_\infty^{\mathrm{can}}$ in [BMCM23, Table 6] and [BCF+25, Table 1].[1] Although the exact direction of the noise ε is unknown, we craft μ such that its evaluation $\mu(\omega)$ at a primitive $(2N)$-th root of unity ω satisfies $|\mu(\omega)| \gg \delta_N \cdot \|\mu\|_\infty$. This ensures that the product $\mu \cdot \varepsilon$ is heavily skewed toward the direction of the ω-evaluation. By repeatedly multiplying by these specially selected μ, the noise grows significantly faster than the heuristic factor $\delta_N \cdot \|\mu\|_\infty$, regardless of the choice of norm, until it exceeds the correctness threshold. The resulting decryption failures leak the most significant bits (MSBs) of the noise, from which the secret key is recovered by linear algebra across all roots of $X^N + 1$, thereby completing the KR-D attack.

The third and last attack exploits the structural dependency of noise growth on the ciphertext distribution during ciphertext-ciphertext (ct-ct) multiplications. In BFV-style multiplications, the dominant noise term $\varepsilon_{\mathrm{ct-ct}}$ is proportional to $t \cdot (\varepsilon_0 \cdot I_1 + \varepsilon_1 \cdot I_0)$, where $I_\beta \approx (a_\beta \cdot s + b_\beta)/q$ represents the rounding error for ciphertexts (a_β, b_β) with underlying noise ε_β for $\beta \in \{0, 1\}$. Standard heuristic bounds often assume both I_β to behave like a random variable with a small norm, whose upper bound is driven by the uniform distributions of a and b (e.g., [BMCM23, Table 6], [BCF+25, Table 1], [CS16, Appendix E.2], or [KPZ21, Section 2.2.2]). However, we demonstrate that an IND-CPA-D adversary can exploit this by maliciously selecting ciphertexts where the polynomial a follows a specific coefficient distribution designed to maximize the expansion of the product $a \cdot s$. For simplicity, we focus on homomorphic squaring where the two input ciphertexts are identical. By requesting multiple fresh encryptions of zero and aggregating them until the a-component satisfies $|a(\omega)| > B$ for a primitive root ω and some threshold $B > 0$, the adversary can control these structural properties to force $\varepsilon_{\mathrm{ct-ct}} \approx 2t \cdot \varepsilon \cdot (a/q) \cdot s$ toward its true worst-case expansion. This effectively defeats the heuristic noise-growth assumptions adopted in major libraries such as OpenFHE [Ope25]. It is what allows us to carry out the addition-based attack of [CCP+24] within the worst-case noise bound: our additions seed a message-dependent noise that a number of squarings then reveal, breaking indistinguishability.

1.3 Other Related Works

Several works have investigated the practical impact of IND-CPA-D attacks. The practicality of attacks against CKKS was investigated in [BCDS+25], when less flooding is used than suggested in [LMSS22]. The applicability of IND-CPA-D attacks to threshold-FHE applications in the federated learning context was exhibited in [BSBK25].

[1] Or similar bound $\|\varepsilon_{\mathrm{cmul}}\|_\infty^{\mathrm{can}} \leq 6\sqrt{N}t \cdot \|\varepsilon\|_\infty^{\mathrm{can}}$ in [GHS12].

The present work focuses on attacks, but let us mention approaches that have been proposed to achieve IND-CPA-D security. In [LMSS22], it was shown that adding noise before decryption thwarts the attack from [LM21] on CKKS. The amount of noise needed was investigated further in [Ogi25]. In [ABMP26], the authors propose designing FHE parameters to enforce correctness for specific circuits; the approach has been recently extended in [BCF+25] for specific BFV settings.

1.3.1 Novelty over prior works. Our attacks uncover critical blind spots that distinguish this work from prior IND-CPA-D analyses [CCP+24, CSBB24]. While Cheon *et al.* [CCP+24] established that noticeable failure probabilities compromise security reductions, their concrete bootstrapping attacks were restricted to TFHE and CKKS, leaving BGV/BFV open. We close this gap, extracting inequality hints from a technically distinct, non-power-of-two cyclotomic regime.

More importantly, our second and third attacks introduce the first dynamic, adaptive evaluation queries targeting homomorphic multiplications. Prior dynamic attempts [CSBB24] were strictly limited to ciphertext noise recovery via dichotomic search on homomorphic additions. By exploiting the geometric shape of ciphertexts rather than just addition noise, our attacks show that widely used heuristic noise estimations continue to rely on assumptions that ignore an adaptive adversary. Consequently, even works explicitly addressing the IND-CPA-D setting in response to recent vulnerabilities, such as recent defenses [Ope25, ABMP26], parameterizations [BMCM23, BCF+25], noise estimation fixes [GZ25], and BGV/BFV analyses [BMMU25], remain vulnerable to our attack surface.

1.4 Code and Experiments

Our code is available at https://github.com/hmchoe0528/NEW_INDCPAD_HE. All experiments were executed single-threaded on an Apple M4 Pro, equipped with a 14-core CPU clocked at 4.41 GHz, 24 GB of unified memory, and running macOS Sequoia 15.7.4.

2 Preliminaries

2.1 Notations

For any positive integer M, we let $[M] := \{1, 2, \ldots, M\}$ denote the set of indices up to M. We define $\mathrm{i} = \sqrt{-1}$, in straight font.

We let $\mathcal{R} = \mathbb{Z}[X]/(\Phi_m(X))$ denote the cyclotomic ring of index m factored as $m = m_1 \cdot \ldots \cdot m_t$ (where the m_i's are pairwise coprime), and $\mathcal{R}_q = \mathbb{Z}_q[X]/(\Phi_m(X))$ for an arbitrary integer $q \geq 2$. Note, when $m = 2N$ is a power of two, it becomes $\mathcal{R} = \mathbb{Z}[X]/(X^N + 1)$.

We will use the L^2-norm $\|\cdot\|_2$ and L^∞-norm $\|\cdot\|_\infty$ of vectors. We may omit the index 2 in the L^2-norm. For a ring element, these norms are taken on its coefficient vector in the powerful basis, which coincides with the usual coefficient vector when $m = 2N$ is a power of two.

Let $Z \sim \mathcal{N}(m, \sigma)$ be the real Gaussian distribution with mean m and standard deviation σ. Let $\mathcal{CN}(m, \sigma)$ denotes the complex Gaussian distribution, where the real and imaginary parts are distributed independently and identically as a real Gaussian distribution with mean $\mathrm{Re}(m)$ and $\mathrm{Im}(m)$, respectively, and standard deviation $\sigma/\sqrt{2}$.

The KL divergence between two distributions D_0 and D_1 is defined as $D_{\mathrm{KL}}(D_0 \| D_1) = \sum_x D_0(x) \ln(D_0(x)/D_1(x))$.

Let $[x]_q$ denote the reduction of x modulo q into $[-q/2, q/2)$, and $\lfloor x \rceil$ the rounding to the nearest integer.

We refer to Appendix A for further preliminaries, including homomorphic operations of specific FHE schemes and probabilistic bounds we will use.

2.2 Fully Homomorphic Encryption

A Fully Homomorphic Encryption (FHE) scheme allows the execution of arbitrary circuits on encrypted data.

Definition 2.1 (Fully Homomorphic Encryption). A fully homomorphic encryption scheme (FHE) is a tuple of efficient algorithms (KeyGen, Enc, Dec, Eval) with the following specifications:

- $\mathsf{KeyGen}(1^\lambda)$ takes as input a security parameter λ, and outputs a secret key sk and a public key pk;
- $\mathsf{Enc}_{\mathsf{pk}}(\mu)$ takes as inputs a public key pk and a plaintext $\mu \in \mathcal{M}$, where $\mathcal{M}$ is a plaintext space, and outputs a ciphertext ct;
- $\mathsf{Eval}_{\mathsf{pk}}(C, (\mathsf{ct}_1, \cdots, \mathsf{ct}_k))$ takes as inputs a public key pk, a circuit C and a tuple of ciphertexts $\mathsf{ct}_1, \ldots, \mathsf{ct}_k$ where k is the number of input wires of C, and outputs a ciphertext ct;
- $\mathsf{Dec}_{\mathsf{sk}}(\mathsf{ct})$ takes as inputs a secret key sk and a ciphertext ct, and outputs a plaintext μ.

For $\varepsilon \geq 0$ and a circuit size bound B, the scheme is said to be (ε, B)-correct if for any pair (sk, pk) output by KeyGen, for any circuit C of size at most B, for any plaintexts $\mu_1, \ldots, \mu_k \in \mathcal{M}$ where k is the number of input wires of C, the following holds with probability $\geq 1 - \varepsilon$ over $\mathsf{ct}_i := \mathsf{Enc}(\mathsf{pk}, \mu_i)$ (for $i \leq k$):

$$\mathsf{Dec}_{\mathsf{sk}}\left(\mathsf{Eval}_{\mathsf{pk}}(C, (\mathsf{ct}_1, \ldots, \mathsf{ct}_k))\right) = C(\mu_1, \ldots, \mu_k) \quad .$$

An FHE scheme is perfectly correct if it is $(0, B)$-correct for any B.

A fundamental security requirement for FHE is indistinguishability under chosen-plaintext attacks (IND-CPA). In the FHE context, IND-CPA security is typically derived from the hardness of the Learning With Errors (LWE) problem or its variants (RLWE, GLWE), alongside circular security assumptions. We define the advantage of an adversary against this notion as follows.

Definition 2.2 (IND-CPA Security). For an FHE scheme (KeyGen, Enc, Dec, Eval), we define the advantage $\mathsf{Adv}^{\mathsf{IND\text{-}CPA}}(\mathcal{A})$ of an adversary $\mathcal{A} = (\mathcal{A}_0, \mathcal{A}_1)$ against the IND-CPA security of Π as:

$$\left| 2 \cdot \Pr\left[\beta = \beta' \;\middle|\; \begin{array}{c} (\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}; \\ \mu_0, \mu_1 \leftarrow \mathcal{A}_0(\mathsf{pk}); \\ \beta \leftarrow \mathcal{U}(\{0,1\}); \\ b' \leftarrow \mathcal{A}_1(\mathsf{pk}, \mathsf{Enc}(\mu_\beta), \mu_0, \mu_1) \end{array} \right] - 1 \right| .$$

The IND-CPA-D security model, introduced in [LM21], extends the IND-CPA notion by granting the adversary $\mathcal{A}$ access to the decryption results of "well-formed" ciphertexts. In addition to modeling situations where the computation result is made public, it is also strictly necessary in the context of threshold-FHE [AJLA+12, BGG+18]. This is formally managed by introducing three stateful oracles $\mathcal{O}_{\mathsf{Enc}}$, $\mathcal{O}_{\mathsf{Eval}}$ and $\mathcal{O}_{\mathsf{Dec}}$, together with a database S that is initially empty. The oracle $\mathcal{O}_{\mathsf{Enc}}$ takes as input a pair of messages (μ_0, μ_1) and returns an encryption of μ_β, where β is the challenge bit as in the IND-CPA game. The oracle $\mathcal{O}_{\mathsf{Eval}}$ takes as input a circuit C and ciphertext indices, and performs homomorphic operations on previously stored ciphertexts. To do so, the challenger maintains a database S of message and ciphertext tuples $(\mu_0, \mu_1, \mathsf{ct})$ created by the oracles, where μ_0 and μ_1 are also computed in plaintext during the execution of $\mathcal{O}_{\mathsf{Eval}}$. The oracle $\mathcal{O}_{\mathsf{Dec}}$ returns a decryption of a ciphertext from the list only if the underlying messages stored are identical (i.e., $\mu_0 = \mu_1$), thereby preventing trivial attacks. The oracles are formally described in Figure 1.

Note that encryption is syntactically performed by the encryption oracle $\mathcal{O}_{\mathsf{Enc}}$ to model the fact that it is run honestly by the adversary (the adversary does not produce ill-formed ciphertexts).

KR-D security, introduced in [LMSS22], is a relaxation of the IND-CPA-D model. Instead of distinguishing between ciphertexts, the adversary's goal is to recover the secret key. Note that IND-CPA-D security implies KR-D security.

Definition 2.3 (IND-CPA-D security). Let (KeyGen, Enc, Dec, Eval) be an FHE scheme. For an adversary $\mathcal{A}$ provided with access to the stateful oracles $\mathcal{O}_{\mathsf{Enc}}$, $\mathcal{O}_{\mathsf{Eval}}$, and $\mathcal{O}_{\mathsf{Dec}}$, we define the advantage $\mathsf{Adv}^{\mathsf{IND\text{-}CPA\text{-}D}}(\mathcal{A})$ against the IND-CPA-D security of Π as:

$$\left| 2 \cdot \Pr\left[\beta = \beta' \;\middle|\; \begin{array}{c} (\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}; \\ \beta \leftarrow \mathcal{U}(\{0,1\}); \\ \beta' \leftarrow \mathcal{A}^{\mathcal{O}_{\mathsf{Enc}}, \mathcal{O}_{\mathsf{Eval}}, \mathcal{O}_{\mathsf{Dec}}}(\mathsf{pk}) \end{array} \right] - 1 \right| .$$

Definition 2.4 (KR-D security). Let (KeyGen, Enc, Dec, Eval) be an FHE scheme. For an adversary $\mathcal{A}$ that is given access to the oracles $\mathcal{O}_{\mathsf{Enc}}$, $\mathcal{O}_{\mathsf{Eval}}$, and $\mathcal{O}_{\mathsf{Dec}}$ with the challenge bit fixed to $\beta = 0$, the success probability $\mathsf{Succ}^{\mathsf{KR\text{-}D}}(\mathcal{A})$ is defined as:

$$\Pr\left[\mathsf{sk} = \mathsf{sk}' \;\middle|\; \begin{array}{c} (\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}; \beta = 0; \\ \mathsf{sk}' \leftarrow \mathcal{A}^{\mathcal{O}_{\mathsf{Enc}}, \mathcal{O}_{\mathsf{Eval}}, \mathcal{O}_{\mathsf{Dec}}}(\mathsf{pk}) \end{array} \right] .$$

The first two attacks we present are KR-D attacks, while the last one is an IND-CPA-D attack.

3 Bootstrapping Attacks for BGV/BFV

In this section, we present key-recovery attacks against the BGV and BFV schemes that exploit bootstrapping failures. In our attack model, the adversary can observe decryption failures that occur after a bootstrap. Such failures introduce structured failures that yield approximate inequality hints on the secret key, which can be leveraged for key recovery. The attacks are inspired by the bootstrapping failure attack on DM/CGGI from [CCP+24]. For the sake of convenience, it is recalled in Appendix A.1.

3.1 Bootstrapping Failures

We recall parts of the bootstrapping procedures for BGV and BFV, and derive correctness conditions, from which we obtain inequalities characterizing (specific) decryption failures.

3.1.1 Focus on the BFV scheme. Let $(a, b) \in \mathcal{R}_q^2$ be a BFV ciphertext encrypting a plaintext $\mu \in \mathcal{R}_{p^r}$ with noise $\varepsilon_{\mathrm{in}}$, i.e.:

$$b + as = \left\lfloor \frac{q}{p^r} \mu \right\rceil + \varepsilon_{\mathrm{in}} \pmod{q} \quad .$$

$\mathcal{O}_{\mathsf{Enc}}(\mu_0, \mu_1; \mathsf{pk}, \beta, i)$
1: $\mathsf{ct} \leftarrow \mathsf{Enc}_{\mathsf{pk}}(\mu_\beta)$
2: $S[i] \leftarrow (\mu_0, \mu_1, \mathsf{ct})$
3: $i \leftarrow i + 1$
4: **return** (ct, i)

$\mathcal{O}_{\mathsf{eval}}(C, i_1, \ldots, i_k; \beta, S)$
1: $\mathsf{ct} \leftarrow \mathsf{Eval}_{\mathsf{pk}}(C, S[i_1].\mathsf{ct}, \ldots, S[i_k].\mathsf{ct})$
2: $r_0 \leftarrow C(S[i_1].\mu_0, \ldots, S[i_k].\mu_0)$
3: $r_1 \leftarrow C(S[i_1].\mu_1, \ldots, S[i_k].\mu_1)$
4: $S[i] \leftarrow (r_0, r_1, \mathsf{ct})$
5: $i \leftarrow i + 1$
6: **return** (ct, i)

$\mathcal{O}_{\mathsf{Dec}}(j; \mathsf{sk}, S)$
1: **if** $S[j].\mu_0 = S[j].\mu_1$ **then**
2: $\quad \mu \leftarrow \mathsf{Dec}_{\mathsf{sk}}(S[j].\mathsf{ct})$
3: $\quad$ **return** μ
4: **else**
5: $\quad$ **return** $\perp$
6: **end if**

Figure 1: Encryption, evaluation and decryption oracles for Definitions 2.3 and 2.4.

Bootstrapping first performs a modulus-switching step that reduces the ciphertext modulus from q to p^e with $e > r$:

$$(\tilde{a}, \tilde{b}) \leftarrow \left(\left\lfloor \frac{p^e \cdot a}{q} \right\rceil, \left\lfloor \frac{p^e \cdot b}{q} \right\rceil \right) \in \mathcal{R}_{p^e}^2 \ ,$$

which satisfies

$$\tilde{b} + \tilde{a}s = p^{e-r}\mu + \tilde{\varepsilon} \pmod{p^e} \ ,$$

for some noise term $\tilde{\varepsilon}$. It then generates a new BFV ciphertext

$$(a', b') \leftarrow \left(\left\lfloor \frac{q \cdot \tilde{a}}{p^e} \right\rceil, \left\lfloor \frac{q \cdot \tilde{b}}{p^e} \right\rceil \right) \in \mathcal{R}_q^2 \ ,$$

which satisfies

$$a' + b's = \left\lfloor \frac{q}{p^e}\left(p^{e-r}\mu + \tilde{\varepsilon}\right) \right\rceil + \varepsilon' \pmod{q} \ ,$$

where $\|\varepsilon'\|_\infty$ is small compared to $\lfloor q/p^e \rceil$. The pair (a', b') is viewed as a BFV ciphertext encrypting $\mu' = p^{e-r}\mu + \tilde{\varepsilon} \in \mathbb{Z}_{p^e}$. Bootstrapping then proceeds by homomorphically computing the rounded division $\mu = \lfloor \mu'/p^{e-r} \rceil \in \mathbb{Z}_{p^r}$, which corresponds to extracting the $e - r$ most significant p-digits, a procedure known as digit extraction. Bootstrapping correctness requires that:

$$\|\tilde{\varepsilon}\|_\infty < \frac{1}{2}p^{e-r} \ .$$

We note that for $p = 2$, the inequality may be taken as non-strict on the left, meaning correctness holds under the condition $\tilde{\varepsilon}_i \in [-p^{e-r}/2, p^{e-r}/2)$ for every i. However, for simplicity of presentation, we continue to use the strict version of the inequality.

Now, let us define the (modulus-switching) rounding noises $\varepsilon_a = \tilde{a} - (p^e/q)\cdot a$ and $\varepsilon_b = \tilde{b} - (p^e/q)\cdot b$. We have $\tilde{\varepsilon} = (p^e/q)\ \varepsilon_{\text{in}} + (\varepsilon_b + \varepsilon_a \cdot s)$, and we conclude that BFV bootstrapping fails when

$$\left\| \left(\varepsilon_b + \varepsilon_a \cdot s\right) + \frac{p^e}{q} \cdot \varepsilon_{\text{in}} \right\|_\infty \geq \frac{p^{e-r}}{2} \ . \tag{1}$$

The first term depends on ε_a, ε_b and s. Assuming that ciphertexts look uniform, one may heuristically assume that the coefficients of both ε_a and ε_b are uniformly distributed in $[-1/2, 1/2)$ and independent of s. The second term represents the ciphertext-specific noise.

3.1.2 Focus on the BGV scheme. Suppose we have a BGV ciphertext $(a, b) \in \mathcal{R}_q^2$ encrypting $\mu \in \mathbb{Z}_{p^r}$ with

$$b + as = \varepsilon_{\text{in}} \pmod{q}, \quad \mu = \varepsilon_{\text{in}} \pmod{p^r} \ .$$

Bootstrapping first performs a modulus-switching to decrease the ciphertext modulus from q to a special modulus $\tilde{q} = p^e + 1$ for some $e > r$. Let us denote $(\tilde{a}, \tilde{b}) \in \mathcal{R}_{\tilde{q}}^2$ the resulting ciphertext, with

$$\tilde{a} = \frac{\tilde{q}}{q}a + p^r\varepsilon_a \ \text{ and } \ \tilde{b} = \frac{\tilde{q}}{q}b + p^r\varepsilon_b \ ,$$

for some rounding noises ε_a and ε_b. Recall that BGV modulus switching preserves the mod-p^r residue of the ciphertext, for correctness, which explains why ε_a and ε_b are multiplied by p^r. Let us write

$$x = \tilde{b} + \tilde{a}s = x_0 + \tilde{q}I \tag{2}$$

over $\mathcal{R}$ (without mod-$\tilde{q}$ reduction), for some $x_0 = [x]_{\tilde{q}}$ satisfying

$$x_0 = \frac{\tilde{q}}{q}\varepsilon_{\text{in}} + p^r\left(\varepsilon_b + \varepsilon_a s\right) \ . \tag{3}$$

Note that $(\tilde{q}/q)\ \varepsilon_{\text{in}}$ is computed over the rationals, but the whole term x_0 still belongs to $\mathcal{R}$. As detailed in Appendix A.2.2, the underlying plaintext is $[x_0]_{p^r} = [(\tilde{q}/q)\ \varepsilon_{\text{in}}]_{p^r} = [(\tilde{q}/q)\ \mu]_{p^r}$, where the division by q is modulo p^r. Then, we have

$$x = x_0 + \tilde{q}I = x_0 + (p^e + 1)I = (x_0 + I) + p^eI \ .$$

A core idea of BGV bootstrapping is that one can recover $x_0 = (x_0 + I) - I \pmod{p^r}$ by recovering both $x_0 + I \pmod{p^r}$ and $I \pmod{p^r}$ from the p-digits of x, if those of $x_0 + I$ and p^eI do not overlap. This requires that[2]

$$\|x_0 + I\|_\infty < \frac{p^e}{2} \ .$$

Bootstrapping then proceeds by recrypting the ciphertext $(\tilde{c}_0, \tilde{c}_1)$ with a large modulus Q, whose underlying plaintext is $x \in \mathbb{Z}_{p^{e+r}}$ with plaintext modulus p^{e+r}, and by extracting the p-digits of the base-p representation of x. Using Equations (2) and (3) leads to

$$x_0 + I = \frac{p^e}{q}\varepsilon_{\text{in}} + \frac{p^{e+r}(\varepsilon_b + \varepsilon_a s)}{\tilde{q}} + \frac{\tilde{b} + \tilde{a}\cdot s}{\tilde{q}} \ . \tag{4}$$

This implies that BGV bootstrapping fails when

$$\left\| \left(\eta_b + \eta_a \cdot s\right) + \frac{p^e}{q}\ \varepsilon_{\text{in}} \right\|_\infty \geq \frac{p^e}{2} \ , \tag{5}$$

for η_a and η_b that can be computed from (a, b) as $\eta_a = (p^{e+r}\varepsilon_a + \tilde{a})/\tilde{q}$ and $\eta_b = (p^{e+r}\varepsilon_b + \tilde{b})/\tilde{q}$.

3.1.3 HElib's optimized BGV bootstrapping. To reduce the number of digits that need to be extracted in the digit-extraction step, HElib introduces the following optimization technique. After modulus-switching to the special modulus $\tilde{q} = p^e + 1$, we obtain $x = \tilde{b} + \tilde{a}\cdot s = (x_0 + I) + p^eI$. At this point, HElib adds multiples of $\tilde{q}$ to the coefficients of the ciphertext $(\tilde{a}, \tilde{b})$, making them divisible by $p^{e'}$ for some $r \leq e' < e$. This gives a new ciphertext $(a', b') = (\tilde{a}, \tilde{b}) + \tilde{q}\cdot(v_a, v_b)$ over $\mathcal{R}^2$, where v_a (resp. v_b) is chosen as $v_a = -\tilde{a} \pmod{p^{e'}}$ (resp. $v_b = -\tilde{b} \pmod{p^{e'}}$). Since $\tilde{q} = 1 \pmod{p^{e'}}$, we

[2] As in the BFV case, the inequality may be taken as non-strict on the left when $p = 2$; however, for simplicity of presentation, we again continue to use the strict version of the inequality.

have $a' = b' = 0 \pmod{p^{e'}}$. Let $y = v_b + v_a s$ and $x' = b' + a's$. We have:

$$x' = x + \tilde{q}y = (x_0 + y + I) + p^e(I + y) \ .$$

Since $x_0 + y + I = 0 \pmod{p^{e'}}$, bootstrapping then proceeds by homomorphically extracting the p-digits of indices $e'+1, \ldots, e$. For this to succeed, it is necessary that

$$\|x_0 + y + I\|_\infty < \frac{p^e}{2} \ .$$

The extra term $y = v_b + v_a \cdot s$ has to be incorporated into the original quantity $x_0 + I$ in Equation (4). We conclude that HElib's BGV bootstrapping fails when $\|x_0 + y + I\|_\infty \geq p^e/2$, which is equivalent to Equation (5) with η_a (resp. η_b) updated to $\eta_a = v_a + (p^{e+r}\varepsilon_a + \tilde{a})/\tilde{q}$ and (resp. $\eta_b = v_b + (p^{e+r}\varepsilon_b + \tilde{b})/\tilde{q}$).

We note that Equations (1) for BFV and (5) for BGV may be viewed as RLWE extensions of Equation (12) for DM/CGGI. In all these schemes, this is the main source of bootstrapping failure.

As a remark, recent unified formulation of BGV and BFV bootstrapping of Geelen and Vercauteren [GV23] uses a different BGV decryption circuit from HElib's; but the resulting correctness condition [GV23, Equation (11)] has the same form as Equation (4), allowing our analysis to be applied as well.

3.2 Failures in the Powerful Basis

So far we have not committed to a specific coordinate representation of the ring elements; we now fix the powerful basis, in which HElib represents them and measures their norm. HElib considers cyclotomic rings whose index is not a power of two, and relies on the powerful representation of ring elements. This representation complicates the recovery of the secret key from bootstrapping failures as a coordinate of s impacts failure via several rounding noise coordinates instead of one.

Recall that $m = m_1 \cdot \ldots \cdot m_t$ where the m_i's are pairwise coprime. The ring $\mathcal{A} = \mathbb{Z}[X_1, \ldots, X_t]/(\Phi_{m_i}(X_i))_{i\in[t]}$ is isomorphic to $\mathcal{R}$ via $\tau : f(X_1, \ldots, X_t) \longmapsto f\left(X^{m/m_1}, \ldots, X^{m/m_t}\right)$. The $\mathbb{Z}$-basis of $\mathcal{A}$, made of all $X_1^{i_1} \cdots X_t^{i_t}$'s for $0 \leq i_j \leq \varphi(m_j),\ j \in [t]$, is called the powerful basis. The *powerful representation* $[w]_{\text{pwfl}} \in \mathbb{Z}^{\phi(m)}$ of an element $w \in \mathcal{R}$ consists of the coefficients of $\tau^{-1}(w)$ in the powerful basis. We will denote the coefficient at index $\mathbf{k} = (k_1, \cdots, k_t)$ of the powerful representation $[w]_{\text{pwfl}}$ as $[w]_{\text{pwfl},\mathbf{k}}$.

Let M_s be the multiplication-by-s matrix in the powerful basis, i.e., for all $w \in \mathbb{R}[X]/\Phi_m(X)$:

$$[w \cdot s]_{\text{pwfl}} = M_s \cdot [w]_{\text{pwfl}} \ ,$$

Assume that $s = \sum_{0\leq i<m} s_i X^i \pmod{\Phi_m(X)}$. Then it was shown in [HS21, Lemma 6.1] that each entry of M_s is the signed sum of 2^t distinct s_i's and that the number of appearances of any s_i in any row of M_s is some 2^ℓ for $\ell \leq t$.[3] The specific structure of M_s, which depends on m, then allows us to reconstruct the inequality hint with a linear combination of observable values (such as η_a, η_b, or later reconstructed to g_0 and c_j's in Section 3.3) with the ternary coefficients s_j.

To perform digit extraction during bootstrapping, the ciphertext must first be transformed from the slot domain to the powerful basis domain via a linear map (SlotToCoeff). In this domain, the homomorphic decryption circuit is evaluated to produce an encrypted value x. The coefficients of x in the powerful basis represent the noisy message components that are subject to the subsequent digit extraction process. Hence, if the magnitude of a coefficient at index $\mathbf{k} = (k_1, \ldots, k_t)$ in the powerful representation exceeds the threshold $p^e/2$, it triggers a wrap-around error, leading to incorrect extraction of digits in the index $\mathbf{k}$. Since each coefficient in the powerful basis is uniquely mapped to a specific slot, this error is lifted to the slots through the final inverse linear transformation (CoeffToSlot). Consequently, the failure appears as a detectable error at the corresponding slot index in the final decrypted result, providing a direct link between the observed failure and the hint from the underlying rounding errors in the powerful basis. When failure occurs at index $\mathbf{k}$, it yields an inequality hint

$$\left|\left[(\eta_b + \eta_a \cdot s) + \frac{p^e}{q}\varepsilon_{\text{in}}\right]_{\text{pwfl},\mathbf{k}}\right| \geq \frac{p^e}{2} \ ,$$

where the index $\mathbf{k}$ refers to the $\mathbf{k}$-th index in the powerful basis.

3.3 Key Recovery from Inequality Hints

We now explain how to recover $s = \sum_{0\leq i<m} s_i \cdot X^i$ from sufficiently many inequality hints. Recall that

$$\eta_a = v_a + \frac{p^{e+r}}{\tilde{q}}\varepsilon_a + \frac{1}{\tilde{q}}\tilde{a} \ \text{ and } \ \eta_b = v_b + \frac{p^{e+r}}{\tilde{q}}\varepsilon_b + \frac{1}{\tilde{q}}\tilde{b} \ .$$

We model each coefficient of v_a, ε_a and $\tilde{a}$ as independent uniform random variables over $[-p^{e'}/2, p^{e'}/2)$, $[-1/2, 1/2)$ and $[-\tilde{q}/2, \tilde{q}/2)$, respectively (and similarly for v_b, ε_b and $\tilde{b}$). The distribution of η_a has zero mean and variance

$$\sigma^2 = \frac{1}{12}\left(p^{2e'} + \frac{p^{2(e+r)}}{\tilde{q}^2} + 1\right) \ .$$

Note that $\eta_a \cdot s$ is the dominant term in Equation (5). Now, fix $\mathbf{k}$ and let the entries of $[\eta_b]_{\text{pwfl}}$ and $[\eta_a \cdot s]_{\text{pwfl}}$ at index $\mathbf{k}$ be denoted by g_0 and g, respectively. As noted in Section 3.2, we have

$$g = \sum_{0\leq k<m} c_k \cdot s_k \ ,$$

where each c_k is a signed sum of multiple entries of $[\eta_a]_{\text{pwfl}}$, and the number of such entries appearing in c_k is a power-of-two integer 2^ℓ for some known $\ell \leq t$. We define

$$Y_j = g_0 + \sum_{k\neq j} c_k \cdot s_k \ .$$

By the Central Limit Theorem, the distribution of Y_j can be approximated by a zero-mean normal law, with

$$\begin{aligned}\mathsf{Var}[Y_j] &= \mathsf{Var}[g_0] + \mathsf{Var}[g] - \mathsf{Var}[c_j \cdot s_j] \\ &= \begin{cases} (1 + 2^t \cdot \tilde{h} \cdot \frac{\phi(m)}{m} - 2^\ell)\,\sigma^2 & \text{if } s_j = \pm 1 \ , \\ (1 + 2^t \cdot \tilde{h} \cdot \frac{\phi(m)}{m})\,\sigma^2 & \text{if } s_j = 0 \ . \end{cases}\end{aligned}$$

Define $\mathbf{c} = (g_0, c_0, \ldots, c_{m-1})$ and $\mathbf{z} = (1, s_0, \ldots, s_{m-1})$. Then, the condition for a bootstrapping failure at the index $\mathbf{k}$ in the powerful basis is

$$\left|\langle \mathbf{c}, \mathbf{z}\rangle + \frac{p^e}{q}\varepsilon_{\text{in},\mathbf{k}}\right| > \frac{p^e}{2} \ ,$$

[3] The number of appearances being a power of two is not borrowed from [HS21, Lemma 6.1] but from its proof, notably [HS21, Equation (7)].

where $\varepsilon_{\text{in},\mathbf{k}}$ is the $\mathbf{k}$-th coefficient of ε_{in} in the powerful basis. We have that $\langle \mathbf{c}, \mathbf{z} \rangle$ is distributed as $c_j \cdot s_j + Y_j$ for each j.

Let $\mathcal{D}^{(\ell)}$ denote the distribution of c_j. It corresponds to the sum of three independent Irwin-hall distributions, each of which is obtained by summing 2^ℓ i.i.d. uniform random variables over $[-p^{e'}/2, p^{e'}/2)$, $[-1/2, 1/2)$ and $[-\tilde{q}/2, \tilde{q}/2)$, respectively. Let $p_0^{(\ell)}$ denote the probability density function of $\mathcal{D}^{(\ell)}$. Then the probability density function of c_j, conditioned on bootstrapping failure (in the $\mathbf{k}$-th powerful-basis coefficient) becomes

$$\begin{cases} p_1^{(\ell)}(x) := \dfrac{\Pr\left[\left|c_j \cdot s_j + Y_j + \frac{p^e}{q}\varepsilon_{\text{in},i}\right| > \frac{p^e}{2} \mid c_j = x\right]}{\Pr\left[\left|\langle \mathbf{c}, \mathbf{z}\rangle + \frac{p^e}{q}\varepsilon_{\text{in},\mathbf{k}}\right| > \frac{p^e}{2}\right]} & \text{if } |s_j| = 1 \ , \\ p_0^{(\ell)}(x) & \text{if } s_j = 0 \ . \end{cases}$$

Based on this, an adversary can examine the distribution of the j-th coefficient of the inequality hint from failing ciphertexts and evaluate its KL divergence from the reference distribution $\mathcal{D}^{(\ell)}$. If $s_j = 0$, the conditional distribution of c_j coincides with $\mathcal{D}^{(\ell)}$, yielding a KL divergence close to 0; otherwise, the failure condition distorts the distribution of c_j, resulting in a larger KL divergence. By the Chernoff-Stein lemma, distinguishing $s_j = 0$ from $|s_j| = 1$ with error probability δ requires about $\ln(1/\delta)/D_{\text{KL}}(p_1^{(\ell)} \| p_0^{(\ell)})$ failing slots. Choosing $\delta = m^{-1}$ to distinguish for all $j \in [m]$, this becomes $\ln m \cdot D_{\text{KL}}(p_1^{(\ell)} \| p_0^{(\ell)})^{-1}$ failing slots.

Identifying the indices j where $s_j = 0$ allows for a significant reduction in the LWE dimension of s, from m to $\tilde{h}$, eventually leading to a full recovery of the secret key by using standard LWE algorithms in a small dimension.

3.4 Experimental Results

We present our implementation results of the key-recovery attack on HElib's BGV bootstrapping.

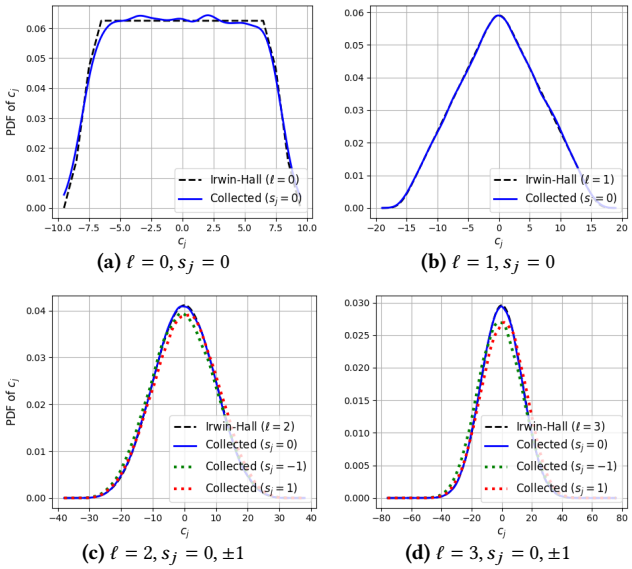
For the sake of efficiency, we focused on the thin bootstrapping variant, for plaintext vectors of dimension smaller than the ring degree. The considered parameters are given in Table 1. All parameter sets in [HS21, Tables 3 and 4] have bootstrapping failure probabilities that are too small for our attack to succeed in a reasonable time (the probabilities are at most of the order of 2^{-80}). For this reason, we modified two parameter sets from [HS21, Table 3],[4] increasing the failure probability to allow the attack to run in reasonable time. We reduced the parameter e from 12 to 8, which increases the probability of per-slot bootstrapping failure to $\approx 5\%$. The Hamming weights of both the decryption key (h) and the bootstrapping secret key ($\tilde{h}$) are set to 120 in [HS21]. For efficient bootstrapping, we set $\tilde{h} = 32$, and we also set $h = 32$. Results with $(h, \tilde{h}) = (120, 32)$ are given in Appendix F.

We ran the attack for parameter sets A* and B*. For parameter set A*, we gathered 492 failing ciphertexts for a total of 23,800 failing slots, which took approximately 45 minutes. Each failed slot gives m values of c_j, corresponding to different values of ℓ. In Figure 2, we present the distributions of the coefficients c_j at position j such that $s_j \in \{-1, 0, 1\}$, obtained from the failed ciphertexts under the parameter set A*. We observed no index j for which $s_j \neq 0$ and c_j can be written as a signed sum of $2^0 = 1$ or $2^1 = 2$ coefficients of $[\eta_a]_{\text{pwfl}}$, explaining the absence of experimental curves for $s_j = \pm 1$ in those cases. We gathered more than 80,000 data points for each graph in Figure 2, except the one in subfigure (a); only 23,800 data points were available for $\ell = 0$. The difference from the Irwin–Hall distributions $D^{(\ell)}$ is quite noticeable when $s_j = \pm 1$. In contrast, we observe that when $s_j = 0$, the coefficients c_j at index j closely match $D^{(\ell)}$.[5] In Appendix C, we additionally present the distributions corresponding to the parameter set B* in Figure 6, which is analogous to Figure 2.

As explained in the previous subsection, the difference between the distributions can be examined using the KL divergence. After sorting, a clear gap appears that almost perfectly distinguishes the indices with non-zero s_j from those with $s_j = 0$. This distinct

Table 1: Considered HElib parameter sets [HS21]. The parameter sets A and B are modified, with modifications highlighted by the → symbols.

	m	$(h, \tilde{h})$	# slots	p^r	(e, e')
A → A*	$257 \cdot 5 \cdot 17$	$(120, 120) \to (32, 32)$	1024	2^1	$(12 \to 8, 4)$
B → B*	$241 \cdot 7 \cdot 17$	$(120, 120) \to (32, 32)$	960	2^1	$(12 \to 8, 4)$



(a) $\ell = 0, s_j = 0$ **(b)** $\ell = 1, s_j = 0$

(c) $\ell = 2, s_j = 0, \pm 1$ **(d)** $\ell = 3, s_j = 0, \pm 1$

Figure 2: Distributions of the coefficients c_j at index j, conditioned on bootstrapping failure, for $\ell \in \{0, 1, 2, 3\}$ and $s_j \in \{-1, 0, 1\}$. Dashed black curves show the PDFs of $\mathcal{D}^{(\ell)}$, while solid blue, dotted green, and dotted red curves show the empirical PDFs of the collected coefficients c_j from failing ciphertexts, where $s_j = 0, -1$, and 1, respectively (parameter set A*).

[4] We selected the first and third ones from [HS21, Table 3] as they are small, hence providing more efficient bootstrapping; we excluded the second one as its cyclotomic order has only two factors instead of three, whereas higher values of ℓ facilitates the distinguishing step of our attack.

[5] We note that the graphs corresponding to nonzero s_j's are asymmetric, which seems to contradict the essentially symmetric probability density functions $p_1^{(\ell)}$ (the term $(p^e/q)\varepsilon_{\text{in},\mathbf{k}}$ being also symmetric); however, since we are working with $p = 2$, the inequality includes an asymmetry (see footnote 2), so that the $p_1^{(\ell)}$'s are, in fact, not symmetric.

Table 2: Bootstrapping attack on the parameter sets A* and B*. The columns show per-slot failure probabilities and real attack cost (including the time for gathering the failing ciphertexts in minutes), with the number of failed ciphertexts and slots. The last row shows the quality of the guess on the nonzero indices of the secret key, on how many correct guesses were included in the $3\tilde{h}+1=97$ candidates.

Parameter	A*			B*		
$\log_2(p_{\text{fail}})$	−4.392			−4.168		
Real ATK cost	28m	32m	36m	48m	58m	69m
#ctxt used	50	150	250	50	200	350
#slots used	2,225	6,948	11,718	2,441	10,265	17,466
#correct guess	7/32	25/32	32/32	8/32	27/32	32/32

separation allows us to pinpoint $\mathcal{O}(\tilde{h})$ candidates for the $\tilde{h}$ nonzero indices, after which lattice reduction can be used to efficiently solve the resulting $\mathcal{O}(\tilde{h})$-dimensional LWE instance. Heuristically, by choosing $\tilde{h}$ outlier indices based on the variances for each $\ell \in [t]$, and adding one additional candidate corresponding to $\ell = 0$, we obtain a set of $(\tilde{h}t+1)$ candidates that contains all $\tilde{h}$ nonzero indices, assuming that a sufficiently large number of failed slots is available. We also remark that $t \le 3$ is commonly used in the literature, so $\tilde{h}t+1$ stays reasonably small. The actual attack costs for the parameters A^* and B^* are reported in Table 2. The parameter sets are claimed to provide more than 80 bits of IND-CPA security [HS21]. The per-slot bootstrapping failure probability p_{fail} is calculated using [HS21, Equation (5)], together with the previously defined standard deviation σ for η_a. The real attacks on the modified parameters show the time taken for gathering the failing slots and categorizing them based on the KL divergence to get the candidate indices for nonzero s_j's. For example, it took 28 (Resp. 32 and 36) minutes to gather 50 (Resp. 150 and 250) failed ciphertexts and 2,225 (Resp. 6,948 and 11,718) failed slots, and then process them to guess non-zero indices. The attack selects $3\tilde{h}+1=97$ indices as candidates, and from roughly 250 failed ciphertexts onward, it successfully recovers all $\tilde{h}=32$ nonzero coordinates of s. The attack then concludes by applying lattice-based techniques on a 97-dimensional LWE instance, obtained by reducing an original $\phi(m)$-dimensional LWE problem with identical modulus and error distribution. We further note that, after accumulating a sufficient number of failed ciphertexts, 492 in our experiment, the indices become easily distinguishable,[6] resulting in 35 candidate positions that include all $\tilde{h}=32$ nonzero coordinates.

We can also estimate the attack cost as follows. Based on the analysis in Section 3.3, distinguishing the s_j's requires about $\ln m \cdot D_{KL}^{-1}$ failed slots, where $D_{KL} := D_{KL}\left(p_1^{(1)} \| p_0^{(1)}\right)$ can be estimated using the probability density functions given there; we use $\ell = 1$ since the unique index with $\ell = 0$ is always kept as a candidate. Consequently, we can estimate the number of required decryption oracle calls, which equals to the required number of thin bootstrappings, by $\#\mathcal{O}_{\text{Dec}} = \ln m \cdot D_{KL}^{-1} \cdot \text{n_slot}^{-1} \cdot {p_{\text{fail}}}^{-1}$, where n_slot is the number of slots used in thin bootstrapping. For the two parameter sets A* and B*, this estimation yields 506 and 611 decryption oracle calls, respectively, more than what was experimentally observed, as no nonzero s_j had $\ell \le 1$ in our experiments. We estimated that for the parameter sets in HElib [HEl23], it requires $O({p_{\text{fail}}}^{-1})$ decryption oracle calls, as expected.

[6] E.g., by visual inspection or simple clustering; see Figures 7 and 8 in Appendix C.

4 Noise Skewing Attack on BFV

The attack we describe in this section targets plaintext-ciphertext (pt-ct) multiplications. It bypasses heuristic noise bounds by carefully selecting plaintexts that induce a noise growth that significantly exceeds these heuristic bounds.

4.1 Attack Description

We first recall the BFV pt-ct homomorphic multiplication and its heuristic noise analysis. We then exploit a gap in the analysis by considering the canonical embedding of the noise, i.e., evaluations in the roots of the defining cyclotomic polynomial. By repeatedly applying pt-ct multiplications with plaintexts having a large entry in its canonical embedding but an L^∞-norm bounded as prescribed, the adversary can legitimately obtain ciphertexts whose noise surpasses the heuristic bound provided by the heuristic analysis. This triggers decryption failures; from the decrypted results, the adversary can learn the secret key. For the sake of simplicity, we assume that $\mathcal{R}$ is a power-of-two cyclotomic ring.

4.1.1 The BFV PT-CT multiplication. Assume that we have a BFV ciphertext $\text{ct} = (a, b) = \text{Enc}(0) \in \mathcal{R}_q^2$. Then we have

$$b + a \cdot s = \varepsilon \bmod q \ ,$$

for some noise $\varepsilon \in \mathcal{R}$. For a plaintext $\mu \in \mathcal{R}_t$, the pt-ct multiplication gives a ciphertext $(a', b') = \mu \cdot (a, b) \bmod q$ in $\mathcal{R}_q^2$, whose noise term becomes $\varepsilon_{\text{cmul}} = \mu \cdot \varepsilon$ unless $\|\mu \cdot e\|_\infty > q/2$. The standard heuristic noise analysis on pt-ct multiplications [BAB+22, KPZ21] states that the inequality

$$\|\varepsilon_{\text{cmul}}\|_\infty \le \delta_N \cdot \|\mu\|_\infty \cdot \|\varepsilon\|_\infty \ \text{ with } \ \delta_N = O(\sqrt{N}) \ , \qquad (6)$$

holds with probability extremely close to 1. Typically, one uses a heuristic $\delta_N = 2\sqrt{N}$ [Ope25, bgvrns-parametergeneration.cpp], [KPZ21, Section 2], [HPS19, Section 6.2]. Repeatedly performing pt-ct multiplications with $\mu_1, \mu_2, \ldots, \mu_K$ results in a ciphertext with noise $\varepsilon_{\text{cmul},K} = \prod_{i\in[K]} \mu_i \cdot \varepsilon$ unless $\left\|\varepsilon_{\text{cmul},K}\right\|_\infty$ exceeds $q/2$, with a heuristic bound

$$\left\|\varepsilon_{\text{cmul},K}\right\|_\infty \le \delta_N^K \cdot \prod_{i\in[K]} \|\mu_i\|_\infty \cdot V_N \ , \qquad (7)$$

where the initial noise is bounded as $\|e\|_\infty \le V_N$.

In BFV, the ciphertext modulus q is set based on a desired number of sequential multiplications, using noise growth formula. In this section, we assume that parameters are set to enable K sequential pt-ct multiplications, using Equation (6) and (7).

4.1.2 Triggering an incorrect decryption. Consider a $(2N)$-th primitive root of unity ω. We choose a plaintext μ for which the ratio $|\mu(\omega)|/\|\mu\|_\infty$ is particularly large, of the order of $\Theta(N)$. For example, the plaintext $\mu = \sum_{0\le j<N} x^j$ satisfies $\|\mu\|_\infty = 1$ and

$$\left|\mu\left(e^{\frac{i\pi}{N}}\right)\right| = \left|\frac{1-e^{\frac{i\pi}{N}\cdot N}}{1-e^{\frac{i\pi}{N}}}\right| = \left|\frac{2}{1-e^{\frac{i\pi}{N}}}\right| = \frac{1}{\sin\left(\frac{\pi}{2N}\right)} \approx \frac{2N}{\pi} \ .$$

We refer to Appendix B for possible choices of μ for each primitive root of unity ω_j, with $\|\mu\|_\infty = 1$ and $|\mu(\omega_j)| \approx 2N/\pi$.

By choosing such plaintexts, we can increase the noise more than heuristically expected, by repeatedly performing pt-ct multiplications:

$$\left\|\varepsilon_{\mathsf{cmul},K}\right\|_\infty \geq \frac{1}{N} \cdot \left|\varepsilon_{\mathsf{cmul},K}(\omega)\right| = \frac{1}{N} \cdot \prod_{i\in[K]} |\mu_i(\omega)| \cdot |\varepsilon(\omega)|$$
$$\approx \frac{1}{N} \cdot \left(\frac{2N}{\pi}\right)^K \cdot \prod_{i\in[K]} \|\mu_i\|_\infty \cdot \|\varepsilon\|_\infty \ ,$$

since $|a(\omega)| \leq N \cdot \|a\|_\infty$ holds for any $a \in \mathcal{R}$. Therefore, the noise after K pt-ct multiplication becomes $\approx (2N/(\pi\delta_N))^K/N \gg 1$ times larger than the heuristic bound obtained using the standard heuristic noise analysis.

The adversary can legitimately perform pt-ct multiplications, as long as the heuristically estimated noise remains below the correct decryption threshold $q/(2t)$. The decryption then provides the most significant bits of the real noise $\varepsilon_{\mathsf{cmul},K}$ as

$$\tilde{\varepsilon}_{\mathsf{cmul},K} := \frac{q}{t} \cdot \left\lfloor \frac{t}{q} \cdot \varepsilon_{\mathsf{cmul},K} \right\rceil \ .$$

Unless $\left\|\varepsilon_{\mathsf{cmul},K}\right\|_\infty > q/2$, the latter gives an approximation up to an absolute noise that is no larger than $q/(2t)$, i.e., we have that $\left\|\tilde{\varepsilon}_{\mathsf{cmul},K} - \varepsilon_{\mathsf{cmul},K}\right\|_\infty \leq q/(2t)$.

4.1.3 Recovering the secret key. From the decrypted results $\tilde{\varepsilon}_{\mathsf{cmul},K}$, one can reveal $\varepsilon_{\mathsf{cmul},K}(\omega)$ accurately as it was designed so that this specific complex embedding is unexpectedly large. Then the attacker can recover an accurate approximation of the phase of the initial noise, since $\varepsilon(\omega) = \varepsilon_{\mathsf{cmul},K}(\omega)/\prod_{i\in[K]} \mu_i(\omega)$ and the μ_i's are known:

$$\varepsilon(\omega) \approx \tilde{\varepsilon}_{\mathsf{cmul},K}(\omega) \Big/ \prod_{i\in[K]} \mu_i(\omega) \ ,$$

with relative precision of $\lesssim (\log_2 q - \log_2 t - 1)$ bits.

By repeating this for every ω, the attacker can obtain approximations of all the phases of ε. If the approximations are sufficiently accurate, the attacker can fully recover the initial noise ε by applying the inverse discrete Fourier Transform (iDFT). Once it knows ε, it can recover s by elementary arithmetic over $\mathcal{R}_q$.

We summarize the attack in Algorithm 1. It can be checked that the attack runs in polynomial time, requiring N non-adaptive decryption queries $\mathcal{O}_{\mathsf{Dec}}$.

Algorithm 1 The BFV PT-CT Multiplication Attack

Input: A fresh encryption $\mathsf{ct} = (a, b) = \mathsf{Enc}(0) \in \mathcal{R}_q^2$ with ring dimension N; a depth K for pt-ct multiplications.
Output: A secret key sk.

for $k \in [N]$ **do**
  $\mathsf{ct}_k \leftarrow \mathsf{ct}$
  for $d \in [K]$ **do**
    $\mathsf{ct}_k \leftarrow \mathsf{CMult}(\mu_k, \mathsf{ct}_k)$ // μ_k chosen as in Appendix B.
  end for
  $\tilde{\varepsilon}_{\mathsf{cmul},K} \leftarrow \mathcal{O}_{\mathsf{Dec}}(\mathsf{ct}_k)$
  $(\hat{\varepsilon}_{\mathsf{obs}})_k \leftarrow \tilde{\varepsilon}_{\mathsf{cmul},K}(\omega_k)/(\mu_k(\omega_k))^K$ // $\omega_k := e^{\mathrm{i}\pi(2k-1)/N}$.
end for
$\varepsilon_{\mathsf{guess}} \leftarrow \lfloor \mathrm{iDFT}(\hat{\varepsilon}_{\mathsf{obs}}) \rceil$
$s_{\mathsf{guess}} \leftarrow a^{-1} \cdot (b - \varepsilon_{\mathsf{guess}})$ in $\mathcal{R}_q$
return $\mathsf{sk} = s_{\mathsf{guess}}$

4.2 Experimental Validation

We first describe noise growth measurements in the sequential pt-ct multiplications, and then present full key-recovery attack results.

4.2.1 Noise growth experiments. We run the attack described in Algorithm 1 using the OpenFHE library. The attack uses K sequential pt-ct multiplications, for a ciphertext modulus set using Equations (6) and (7). We use a high-probability bound $V_N := B_{\mathsf{err}} \cdot (1 + 2\sqrt{N})$, following the default OpenFHE setting, where B_{err} is a bound for fresh Gaussian noises.

We set $\delta_N = 2\sqrt{N}$. We also set the maximum legitimate number of multiplications K with respect to any plaintext in $\mathcal{R}_t$, relying on the bound $\|\mu\|_\infty \leq t/2$. From Equation (7), the maximum pt-ct depth K and ciphertext modulus q then need to satisfy

$$\frac{q}{2t} > \left(\delta_N \cdot \frac{t}{2}\right)^K \cdot V_N \ .$$

We first present the heuristic behavior of noise ε_d after d pt-ct multiplications, in Figure 3. It shows the real and estimated noise growth with respect to the above equation, for $N = 2^{16}$, $q \approx 2^{240}$ and $t = 1{,}048{,}577 \approx 2^{20}$, with $|\mu(\omega_i)| \approx 2^{33.2}$ and $\|\mu\|_\infty \approx 2^{18.9} \lesssim t/2$. The figure plots the noise magnitudes in $\log_2$-scale, illustrating the exponential growth for different norms. The triangles (▲) represent the $\log_2$ value of the L^∞-bound of ε_d estimated based on Equations (6) and (7): it exhibits a steady linear increase. The squares (■) and the diamonds (♦) denote the measured magnitudes $\log_2 |\varepsilon_d(\omega)|$ and $\log_2 \|\varepsilon_d\|_\infty$ during the attack, respectively, where $\omega = \omega_k = \exp((2k-1)\pi\mathrm{i}/N)$. As expected, the quantity $\log_2 |\varepsilon_d(\omega)|$ increases linearly with a slope of $\log_2 |\mu(\omega)|$, resulting in a steeper trajectory than the estimated L^∞-bound. Regarding the L^∞-norm $\|\varepsilon_d\|_\infty$, while its initial growth is slower than that of $|\varepsilon_d(\omega)|$, it quickly transitions into an exponential increase that closely tracks $|\varepsilon_d(\omega)|$, after only a few multiplications. This behavior arises from the alignment of ε_d with μ (which possesses an extreme phase) after the initial iterations. Then, the growth of the L^∞-norm becomes proportional to the increase in its evaluation at ω, with $\|\varepsilon_d \cdot \mu\|_\infty / \|\varepsilon_d\|_\infty \approx \mu(\omega)$.

This accelerated growth affects the IND-CPA-D security of BFV. Specifically, decryption is allowed only as long as the L^∞-bound of the error remains below the threshold $q/(2t)$. Based on standard heuristic estimates, our choice of μ should theoretically allow up to 7 sequential multiplications. However, we observe that the 7-th multiplication results in $\|\varepsilon_7\|_\infty \gg q/(2t)$. Consequently, a "legitimate" decryption at this moment leaks the Most Significant Bits (MSBs) of ε_7, directly exposing an approximation of the initial noise $\varepsilon_0(\omega)$.

Another numerical example is given in Appendix D.

4.2.2 Key recovery experiments. The behavior of the noise leads to the full secret key recovery attack. In Table 3, we summarize the attack cost for the selected parameters of BFV, where the parameters are generated using the parameter generation tools of the OpenFHE library. Note that most libraries, including OpenFHE, do not provide

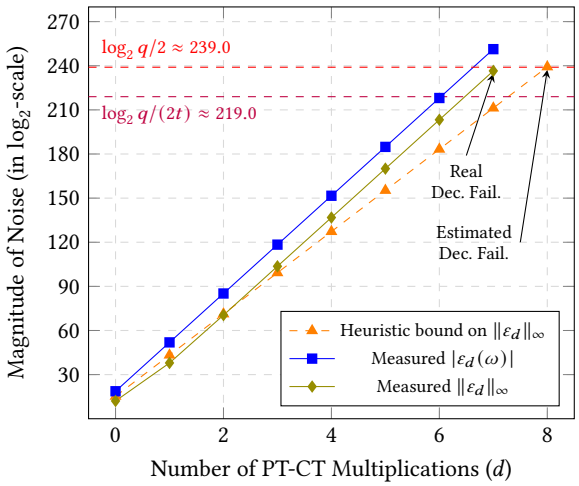


Figure 3: Noise Growth in Algorithm 1; $N = 2^{16}$, $q \approx 2^{240}$, $t \approx 2^{20}$, index $k = 7$, $\|\mu\|_\infty \approx 2^{18.9}$, and $|\mu(\omega)| \approx 2^{33.2}$.

Table 3: BFV PT-CT KR-D Attack Results. The parameter sets provide 128 bits of IND-CPA security, except for $log_2 N = 12$ and 13. $\log_2 q \approx 240$ is used for all parameters. The plaintexts used in the attacks are presented both via their L^∞-norm and through their evaluations at ω. The symbols d_{leg} and d_{atk} denote the legitimate number of pt-ct multiplications and the number of multiplications used in the attack, respectively. The number of decryption oracle queries is denoted by #Dec. The symbols T_c and T_s respectively denote the time taken to collect $\hat{\varepsilon}_{\text{obs}}$ via multiplications and decryptions, and to recover the secret key sk from the decrypted results.

N	t	$\|\mu\|_\infty$ (in $\log_2$)	$\|\mu(\omega)\|$	d_{leg}	d_{atk}	#Dec	T_c	T_s
12	19.6	18.6	28.9	8	8	N	34.1s	0.09s
13	22	21	32.4	7	7	N	108s	0.35s
14	26	25	37.4	6	6	N	367s	1.37s
15	25	24	37.3	6	6	N	1,824s	5.50s
16	20	18.9	33.2	7	7	N	6,834s	21.5s

noise estimation for pt-ct multiplications, so we independently set the legitimate number of pt-ct multiplications.

As shown in Table 3, the attack in Algorithm 1 *perfectly recovers the uniform ternary secret* within N decryption oracle calls (one for each root of $X^N + 1$), staying within the heuristic bounds.

5 Ciphertext Skewing Attack on BFV

This attack targets ciphertext-ciphertext (ct-ct) multiplications. It invalidates heuristic noise analyses by manipulating the ciphertext distribution, which is typically assumed to be uniform distributed modulo the ciphertext modulus. Specifically, the adversary "skews" the ciphertext distribution to increase the quotient of the modular reduction in the decryption equation in well-chosen specific directions. The skewed ciphertext is then used in a ciphertext-ciphertext multiplication, where it triggers a noise that surpasses the heuristic noise upper bound.

5.1 The BFV CT-CT Multiplication Attack

The first component of the attack is to find an encryption of 0 to add to an input ciphertext, so that the sum has an a-part that is unexpectedly large when evaluated in a root ω of the defining polynomial $X^N + 1$ (recall that N is a power of two). This leads to a large quotient I modulo the ciphertext modulus q, in the decryption equation. This ciphertext then induces a noise growth in a subsequent ct-ct multiplication, which surpasses the heuristic noise growth bound. Repeating the process a few times eventually leads to a decryption noise that can be exploited to recover the secret key.

5.1.1 Analysis of CT-CT multiplication. Assume that we have a BFV ciphertext $\text{ct} = (a, b) = \text{Enc}(0) \in \mathcal{R}_q^2$ with a scale factor $\Delta = \lfloor q/t \rfloor$. Then we have $b + a \cdot s = \varepsilon + qI$ for some quotient $I \in \mathcal{R}$ and some noise $\varepsilon \in \mathcal{R}$. When performing a ct-ct multiplication of ct with itself (to simplify), we have the resulting noise

$$\varepsilon_{\text{sqr}} = \frac{t}{q} \cdot \varepsilon^2 + 2t \cdot \varepsilon \cdot I \ .$$

Since $\|\varepsilon\|_\infty \ll q$, the dominating term is $2t \cdot \varepsilon \cdot I$. In heuristic noise estimates, it is assumed that, for some $\delta_N = O(\sqrt{N})$, we have

$$\left\|\varepsilon_{\text{sqr}}\right\|_\infty \approx 2t \cdot \|\varepsilon \cdot I\|_\infty \lesssim 2t\delta_N \cdot \|\varepsilon\|_\infty \cdot \|I\|_\infty \ ,$$

with probability extremely close to 1 under the randomness of ε and/or a. Considering that $qI = b + a \cdot s - \varepsilon \approx a \cdot s \in \mathcal{R}$, one can also bound I as $\|I\|_\infty \lesssim \delta_N \cdot \|a/q\|_\infty \cdot \|s\|_\infty = \delta_N/2$, leading to

$$\left\|\varepsilon_{\text{sqr}}\right\|_\infty \lesssim t\delta_N^2 \cdot \|\varepsilon\|_\infty \ , \tag{8}$$

with overwhelming probability.

In practice, following [BAB+22, KPZ21], an overwhelming-probability L^∞-bound for the noise after K sequential homomorphic multiplications is derived by setting $\delta_N = 2\sqrt{N}$. To ensure correctness of decryption, the following condition must be satisfied:

$$\frac{q}{2t} \gtrsim (4Nt)^K \cdot V_N \ ,$$

where V_N represents a high-probability L^∞-bound on the initial noise, and the factor 2 serves as a safety margin. We will focus on that choice of δ_N in this section.[7]

5.1.2 Beating the heuristic noise bound. The heuristic bound $\|\varepsilon \cdot I\|_\infty \le \delta_N \cdot \|\varepsilon\|_\infty \cdot \|I\|_\infty$ with $\delta_N = O(\sqrt{N})$ is derived from the assumption that ε or I is random and symmetric, producing cancelations in sums of terms. If an adversary can skew the distribution of ε or I, the heuristic noise analysis may no longer hold, possibly opening a way to a successful attack.

For a given ciphertext ct, we first find an encryption of 0 that causes the I-part of $\text{ct} + \text{Enc}(0)$ to have a large phase $I(\omega)$. Then, we apply homomorphic squaring to outpace the heuristic noise

[7] Note that a recent update to the OpenFHE library (v1.4.2.4, #1007) revised this heuristic bound to approximately $(8Nt)^K \cdot V_N$. This change stems from a more conservative estimation of the rounding error I, where $\|I\|_\infty$ is bounded by a distinct extension factor $\delta_{\text{ms},N} = 4\sqrt{N}$ instead of general $\delta_N = 2\sqrt{N}$. While this stricter bound increases the practical complexity of our attack, the underlying rationale for the noise growth analysis remains applicable. For e.g., $k_a = 7.81$ and $k_s = 3.40$ allow exceeding the bound within $\approx 2^{93.5}$ trials for $N = 2^{16}$; we refer to Appendix E.3.

bound. Adding an encryption of 0 also adds an independent noise to the noise of ct. Initially, the two noise terms have a similar magnitude, so this sum changes the noise distribution non-negligibly; however, when we repeat the process iteratively, the impact of the subsequently used encryptions of 0 vanishes, as the noise in the ciphertext undergoing repeated squarings grows.

Let us delve into the details. Our input $\mathrm{ct} = (a, b)$ satisfies $b + a \cdot s = \varepsilon_0 + q \cdot I$. Consider a phase given by a $(2N)$-th primitive root of unity ω. To make $|I(\omega)|$ large, it suffices to make $|a(\omega)|$ large, since

$$I(\omega) = \left\lfloor \frac{b + as}{q} \right\rceil (\omega) \approx \frac{a}{q}(\omega) \cdot s(\omega) \ .$$

Following Appendix A.3, we heuristically assume that set $s(\omega) \sim \mathcal{CN}(0, 2N/3)$ and $(a/q)(\omega) \sim \mathcal{CN}(0, N/12)$, and we have

$$\Pr\left[\left|\frac{a}{q}(\omega)\right| > k_a \cdot \sqrt{\frac{N}{12}}\right] = \exp\left(-k_a^2\right) \ ,$$

$$\Pr\left[|s(\omega)| > k_s \cdot \sqrt{\frac{2N}{3}}\right] = \exp\left(-k_s^2\right) \ ,$$

for all $k_a, k_s > 0$, which gives

$$\begin{aligned}\left|\varepsilon_{\mathrm{sqr}}(\omega)\right| \approx 2t \cdot |I(\omega)| \cdot |\varepsilon(\omega)| &\gtrsim 2t \cdot k_a\sqrt{\frac{N}{12}} \cdot k_s\sqrt{\frac{2N}{3}} \cdot |\varepsilon(\omega)| \\ &= \left(\frac{\sqrt{2}}{3} k_a k_s\right) \cdot Nt \cdot |\varepsilon(\omega)| \ ,\end{aligned}$$

with probability $\exp(-k_a^2 - k_s^2)$. For N possible choices of ω, the probability grows by a factor of $N/2$ (halved due to the conjugacy). This increase is larger than that obtained by instantiating Equation (8) with $\delta_N = 2\sqrt{N}$ when

$$k_a k_s > 6\sqrt{2} \ . \tag{9}$$

Note that the comparison is not completely fair, as the above considers evaluation in ω while Equation (8) considers the infinity norm; however, repeating the process will make the difference of norms less relevant. This implies that once $|s(\omega)| = k_s \cdot \sqrt{2N/3}$ is sufficiently large for some ω, one can exceed the heuristic noise growth $4Nt$ by selecting a's whose $(a/q)(\omega)$ is larger than $k_a\sqrt{N/12}$, when $k_a > 6\sqrt{2}/k_s$.

We stress that even though a is defined as an element of $\mathcal{R}_q$, for the purpose of understanding I, it is viewed as an element of $\mathcal{R}$ whose coefficients belong to $[-q/2, q/2]$. Given the above, for fresh encryptions $\mathrm{ct}^{(j)} \leftarrow \mathsf{Enc}(0)$, we can try to find $\mathrm{ct}_1 \leftarrow \mathrm{ct}_0 + \mathrm{ct}^{(j_1)}$ which has an a-component a^* with an unexpectedly large $|a^*(\omega)|$, by brute-force search (it is computationally realistic as the maximization is 1-dimensional). Finding $\mathrm{ct}^{(j_1)}$ that gives a large $|a^*(\omega)|$ will require a number of encryption queries, depending on the magnitude threshold. After homomorphic squaring $\mathrm{ct}_{\mathrm{sqr},1} \leftarrow \mathsf{Mult}(\mathrm{ct}_1, \mathrm{ct}_1)$, the noise amplifies as

$$\varepsilon_{\mathrm{sqr},1}(\omega) \approx 2t \cdot \varepsilon(\omega) \cdot I(\omega) \approx 2t \cdot \varepsilon(\omega) \cdot \frac{a^*(\omega)}{q} \cdot s(\omega) \ ,$$

where ε is the noise in $\mathrm{ct}_1 = \mathrm{ct}_0 + \mathrm{ct}^{(j_1)}$.

We continue by finding $\mathrm{ct}^{(j_2)} \leftarrow \mathsf{Enc}(0)$ that makes the a-component of $\mathrm{ct}_2 \leftarrow \mathrm{ct}_{\mathrm{sqr},1} + \mathrm{ct}^{(j_2)}$ large with respect to ω, then squaring homomorphically as $\mathrm{ct}_{\mathrm{sqr},2} \leftarrow \mathsf{Mult}(\mathrm{ct}_2, \mathrm{ct}_2)$. Note that the fresh encryption noise becomes relatively small compared to $\varepsilon_{\mathrm{sqr},1}$.

After K repetitions, we obtain a ciphertext $\mathrm{ct}_{\mathrm{sqr},K} = \mathrm{ct}_K$ with the noise $\varepsilon_{\mathrm{sqr},K} = \varepsilon_K$ satisfying

$$\varepsilon_K(\omega) \approx \left(2t \cdot s(\omega)\right)^K \cdot \left(\prod_{i \in [K]} \frac{a_i^*(\omega)}{q}\right) \cdot \varepsilon(\omega) \ ,$$

where a_i^* is the a-part of the ciphertext $\mathrm{ct}_i \leftarrow \mathrm{ct}_{\mathrm{sqr},i-1} + \mathrm{ct}^{(j_i)}$ for the encryption $\mathrm{ct}^{(j_i)}$ of 0 selected in the i-th iteration.

Note that one can reduce the number of encryption queries to find an appropriate $\mathrm{ct}^{(j_i)}$ by linearly combining them. Furthermore, from the second iteration, we can reuse the already obtained fresh encryptions $\{\mathrm{ct}^{(j)}\}_{j \neq j_1, \ldots, j_{i-1}}$, or their linear combinations. These strategies decrease the number of calls to the encryption oracle.

For sufficiently large values of $|s(\omega)|$, $|(a_i^*/q)(\omega)|$ (for all $i \in [K]$), and $|\varepsilon(\omega)|$, the resulting large noise phase $|\varepsilon_{\mathrm{sqr},K}(\omega)|$ may cause $\left\|\varepsilon_{\mathrm{sqr},K}\right\|_\infty$ to exceed both the heuristic upper bound on the noise and the decryption threshold. In this case, decryption fails and produces incorrect plaintext.

5.1.3 Decryption failures and IND-CPA-D attack. We now show that the capability to induce decryption failures can be leveraged to mount an IND-CPA-D attack, extending the addition-based BFV attack of [CCP+24]. The core insight is that the adversary can build, out of the challenge ciphertexts, a ciphertext whose tracked plaintext is 0, hence legitimately decryptable. The noise, however, carries a deterministic term depending on the challenge message, which is then amplified by the previously described methodology. The decrypted result then provides a distinguishing hint.

We consider an adversary $\mathcal{A}$ in an IND-CPA-D setting. For a challenge bit $\beta \leftarrow \{0, 1\}$, the challenger will provide ciphertexts $\mathsf{Enc}(\mu_\beta)$ once the adversary queries $\mathcal{O}_{\mathsf{Enc}}$ on two messages μ_0, μ_1. The adversary $\mathcal{A}$ proceeds as follows, with parameter k_a:

Phase 1: Forcing Failure and Selecting ω. The adversary $\mathcal{A}$ identifies a $(2N)$-th primitive root of unity ω that leads to a concrete decryption failure for an encryption of 0.

- It first picks ω randomly and queries $\mathcal{O}_{\mathsf{Enc}}$ for $\mu_0 = \mu_1 = 0$ to find a ciphertext ct_0 whose a-component satisfies $|a(\omega)| > k_a\sqrt{N/12}$.
- It then performs the iterative K-squaring process as previously described, to generate ct_K, and queries $\mathcal{O}_{\mathsf{Dec}}(\mathrm{ct}_K)$. Since ct_K is tracked as an encryption of 0 in the shared state, this query is permitted. If decryption succeeds without failure, it restarts the process with a different ω.
- Once a failure is confirmed, $\omega_{\mathrm{succ}} := \omega$ is retained. This specific ω_{succ} is re-used in Phases 2 and 3.

Phase 2: Reference Distribution Building. With ω_{succ} prepared, the adversary $\mathcal{A}$ runs the distinguishability experiments with the challenge messages $\mu_0 = 0$ and $\mu_1 = \mathcal{H}$, where $\mathcal{H}$ is built in Equation (10) below.

- To build reference distributions, it queries $\mathcal{O}_{\mathsf{Enc}}$ on the pair (μ_γ, μ_γ) a total of M times, obtaining $\mathrm{ct}_{\gamma,1}, \ldots, \mathrm{ct}_{\gamma,M}$, for $\gamma \in \{0, 1\}$.

- It then computes $\mathrm{ct}_\gamma = \sum_{i\in[M]} c_i \cdot \mathrm{ct}_{\gamma,i}$ with integer coefficients c_i whose sum $C := \sum_i c_i$ is a multiple of t, using ciphertext additions only. Its tracked plaintext is $C \cdot \mu_\gamma = 0 \in \mathcal{R}_t$.
- It then runs the K-fold amplification of Phase 1, but only with ω_{succ}, on ct_γ and queries $\mathcal{O}_{\mathrm{Dec}}$ on the result, whose tracked plaintexts remain 0.
- By repeating this, it obtains an empirical distribution $\mathcal{D}_\gamma$ of the decryption results, for $\gamma \in \{0,1\}$.

Phase 3: Challenge Analysis. Finally, the adversary $\mathcal{A}$ submits $\mu_0 = 0$ and $\mu_1 = \mathcal{H}$ as the challenge messages and receives M ciphertexts $\{\mathrm{ct}_i^* = \mathsf{Enc}(\mu_\beta)\}_{i=1..M}$ for a challenge bit β.

- It then computes $\mathrm{ct}^{\#} = \sum_{i\in[M]} c_i \cdot \mathrm{ct}_i^*$ and the K-fold amplification, as in Phase 2.
- It then queries $\mathcal{O}_{\mathrm{Dec}}$ on the resulting ciphertext.
- By repeating this, it obtains an empirical distribution $\mathcal{D}^{\#}$, and guesses β' by comparing it with $\mathcal{D}_0$ and $\mathcal{D}_1$.

The success of this attack relies on three main aspects: 1) the existence of an $\omega = \omega_{\mathrm{succ}}$ that makes $|s(\omega_{\mathrm{succ}})|$ sufficiently large to pass Phase 1 (which will then make the later phases work); 2) whether the two distributions $\mathcal{D}_0$ and $\mathcal{D}_1$ are efficiently distinguishable; and 3) whether the whole circuit is still allowed under the noise analysis.

The first can be guaranteed by setting k_a (and k_s) appropriately. For instance, with a probability of $\mathcal{O}(1)$, there exists at least one ω that makes $|s(\omega)| \gtrsim k_s \cdot \sqrt{2N/3}$ for $k_s = \sqrt{\ln(N/2)}$ (see Appendix A). Hence, we can set $k_a > 6\sqrt{2}/\sqrt{\ln(N/2)}$ to trigger a decryption failure for ct_K.

For the second, we need the message to leave a trace in the noise. We choose the challenge plaintext $\mathcal{H}$ to make this happen. Let us stick with the RNS-based instantiation of BFV, for e.g., as in OpenFHE [BAB+22, KPZ21]: For an encoding map $\Delta : \mathcal{R}_t \to \mathcal{R}$, a ciphertext of $\mu \in \mathcal{R}_t$ has phase $\varphi := b + as = \Delta(\mu) + \varepsilon + qI$ over $\mathcal{R}$. By letting

$$\nu(\mu) := q \cdot \mu - t \cdot \Delta(\mu) \in \mathcal{R} \ ,$$

we have $\|\nu(\mu)\|_\infty < t$ and, coefficient-wise, $\nu(\mu)_j \equiv q \cdot \mu_j \pmod t$. Then, we let

$$\mathcal{H} := \sum_{0\le j<N} \mathrm{sign}\left(\cos\left(\pi \frac{(2k-1)(2j+1)}{2N}\right)\right) \cdot m^\star \cdot X^j \ , \qquad (10)$$

which is the plaintext of Equation (18) multiplied by $m^\star$, where $m^\star := \mathrm{argmax}_{0<m\le t/2}[qm]_t$. Since $\gcd(q,t) = 1$, $[qm^\star]_t$ reaches near t, and thus, $|\nu(\mathcal{H})(\omega)| = [qm^\star]_t/\sin(\pi/(2N))$ reaches near the maximum $\approx 2Nt/\pi$. Note that $\nu(0) = 0$.

Now let $\mathrm{ct}_1, \ldots, \mathrm{ct}_M$ all encrypt μ, with noises $\varepsilon_1, \ldots, \varepsilon_M$, and let $c_1, \ldots, c_M$ be integers whose sum $C := \sum_i c_i$ is a multiple of t. The ciphertext $\mathrm{ct} := \sum_i c_i \mathrm{ct}_i$, obtained with ciphertext additions only, has tracked plaintext $C \cdot \mu = 0 \in \mathcal{R}_t$ with noise

$$\varepsilon = \sum_i c_i \cdot \varepsilon_i - \rho \cdot \nu(\mu) \ , \qquad (11)$$

where $\rho := C/t$. Assuming independence and randomness on ε_i's, the ratio between the two terms, $|(\sum_i c_i\varepsilon_i)(\omega)|$ and $|(\rho \cdot \nu(\mathcal{H}))(\omega)|$, approximates to

$$\frac{\rho\,|\nu(\mathcal{H})(\omega)|}{\sqrt{\sum_i c_i^2} \cdot |\varepsilon_{\mathsf{fresh}}(\omega)|} \gtrsim \frac{C/t \cdot 2Nt/\pi}{C/\sqrt{M} \cdot \sqrt{N} \cdot V_N} \approx \sqrt{M} \cdot \frac{2\sqrt{N}}{\pi V_N} \ ,$$

where c_i's are set to have similar magnitudes. Since the squaring iteration is linear in the noise, this ratio is maintained even after the K-fold squarings, making the two distributions $\mathcal{D}_0$ and $\mathcal{D}_1$ distinguishable for sufficiently large M.

For the third, the additions of Phases 2 and 3 leave the noise after additions bounded by $C \cdot V_N + C \approx C \cdot V_N$ from Equation (11), where V_N is the high-probability bound on the noise of a fresh ciphertext. This worst-case multiplicative factor C can be taken into account in the noise bound; the circuit used in the attack is therefore admissible whenever

$$(4Nt)^K \cdot C \cdot V_N < \frac{q}{2t} \ .$$

5.2 Experimental Validation

We use OpenFHE [Ope25], whose noise estimate predates the revision discussed in the footnote 7.[8]

We first present the noise growth obtained during Phase 1 of the attack, in Figure 4, with $\log_2 N = 16$, $\log_2 q \approx 1{,}352$ with $\log_2 P \approx 360$, and $t = 786{,}433$, allowing 35 depths of multiplications. The tails for $|s(\omega)| \approx k_s\sqrt{2N/3}$ and $|a(\omega)| \approx k_a\sqrt{N/12}$ were $k_s = 3.31$ and $k_a = 4.5$, respectively; such an ω exists for a random secret key with a probability very close to 1. For enlarging the a-components, we require an order of $\exp(k_a^2) \approx 2^{29.2}$ candidates to be tested; we simply simulated the outcome using the secret key instead of searching by brute force, drawing a uniformly from $\mathcal{R}_q$ and shifting it at ω alone, so that $\|a/q\|_\infty$ and the remaining coordinates stay as they would be for genuine encryptions of 0.

We also ran the IND-CPA-D attack of Phases 2 (and 3) with the same parameters, using $M = 2^{18}$ encryption queries per run, $\rho = 24{,}800$, $K = 34$, and an ω_{succ} with $k_s = 3.31$. For this parameter set, $m^\star = 321{,}568$ and $[qm^\star]_t = t - 3$, so that $|\nu(\mathcal{H})(\omega)| \approx 2^{34.93}$. With $V_N = B_{\mathrm{err}}(1+2\delta_N) \approx 2^{14.26}$ and $C = \rho \cdot t \approx 2^{34.18}$, the estimated bound is given as,

$$(4Nt)^K \cdot C \cdot V_N \approx 2^{1326.3} < \frac{q}{2t} \approx 2^{1331.4} \ .$$

Figure 5 reports 40 runs for each of $\gamma \in \{0,1\}$. The largest returned coefficient averages 4 for $\mathcal{D}_0$ against 90 for $\mathcal{D}_1$, which can be straightforwardly distinguished.

In the end, Phase 1 required around $\min\big(N/2, \exp(k_s^2)\big) \approx 2^{15}$ decryption queries (one per candidate ω, of which there are $N/2$ up to conjugacy) and around $\exp(k_a^2) \approx 2^{29}$ candidate a-components to be tested at each of its squarings. Since the a-components are all different, a single $\exp(k_a^2)$-pool serves for every squaring. The number of tests (to get large a-components) is $K \cdot \exp(k_a^2) \approx 2^{34}$ per run. We note that the pool itself need not hold $\exp(k_a^2)$ ciphertexts, since the subset sums of some encryptions of 0 again form encryptions of 0, while the noise they carry is negligible against the noise carried in Phases 2 and 3.

Further experiments are provided in Appendix E.

6 Conclusion

This work reaffirms the central role of *correctness* in the IND-CPA-D security of exact FHE schemes: its failure directly breaks the simulation-based security of any Threshold FHE built on them for

[8] We provide, in Appendix E.3, the corresponding figures for the later release [Ope26], to which the attack also applies.

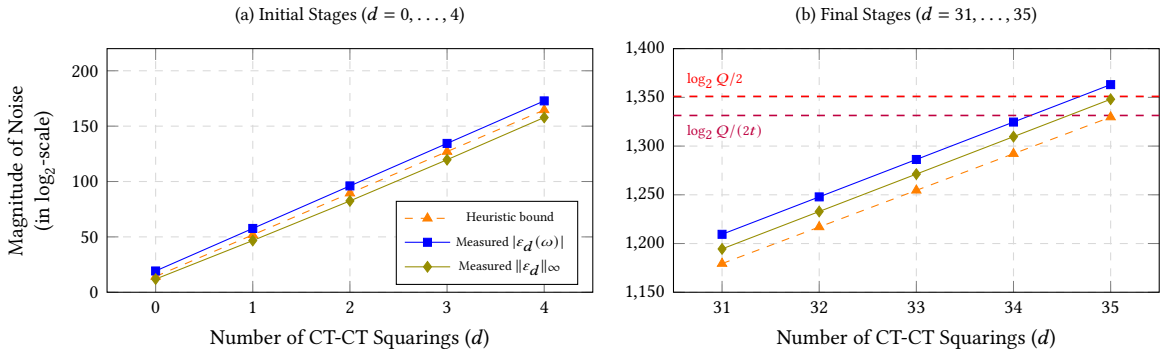


Figure 4: Noise Growth in BFV CT-CT Squarings (initial and final stages). Parameters are $N = 2^{16}$, $\log_2 Q = 1352$, $\log_2 P = 360$, and $t = 786{,}433$. We choose ω such that $k_s = 3.31$. The full trend can be found in Figure 10.

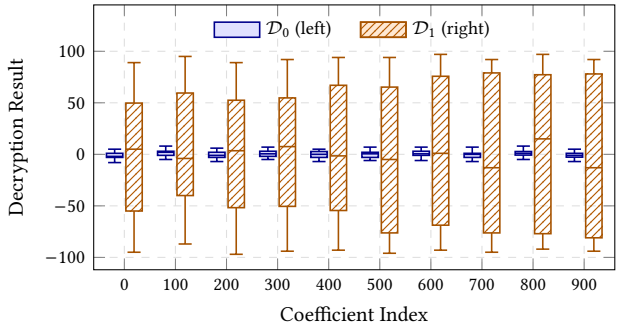


Figure 5: Empirical distributions $\mathcal{D}_0$ (left of each pair, solid) and $\mathcal{D}_1$ (right, hatched) of the decryption results, at coefficient indices $0, 100, \ldots, 900$, over 40 runs each. $N = 2^{16}$, $q \approx 2^{1352}$ with $P \approx 2^{360}$, $t = 786{,}433$, $k_a = 4.5$, $k_s = 3.31$, $M = 2^{18}$, $\rho = 24{,}800$, $K = 34$. Both are centred at 0, but aparted by their *scale*. The inter-quartile range differs by a factor of 31.6 on average.

MPC. Whereas prior IND-CPA-D cryptanalyses largely questioned the soundness of noise bounds in their statistical form, ours is, to our knowledge, the first to *outpace* these bounds not by sharper probabilistic accounting but by crafting evaluation circuits whose inputs and intermediate ciphertexts bypass the independence assumptions underlying current noise analyses. The takeaway is that probabilistic savings and variance-based estimates, accurate on average under IND-CPA, are not an adequate basis for parameter selection under IND-CPA-D: correctness must be argued against adversaries who craft the evaluation circuit, not merely against typical inputs.

6.1 Applicability and Extendability

Our insights generalize beyond the schemes we focus on. The first attack applies to BFV bootstrapping as well; we chose BGV for its widely used bootstrapping implementation and more challenging algebraic structure. Although evaluated on sparse ternary secrets, it extends to larger sparse secrets by generalizing the conditional densities $p_0^{(\ell)}, p_1^{(\ell)}$ of c_j to further boundaries $p_2^{(\ell)}, p_3^{(\ell)}, \ldots$ for coefficients $|s_j| \geq 2$.

The second and third attacks remain viable across secret distributions: concrete costs depend on the corresponding noise analysis, but the gap between average-case and worst-case noise stays exploitable under current heuristics. They do not, however, translate to BGV, where every multiplication is immediately followed by a modulus switching that is absent in BFV; bypassing this by scheme-switching is disallowed within the BGV IND-CPA-D game. Adapting these attacks to BGV remains open.

6.2 Countermeasures

We propose countermeasures for each attack. For the first, bootstrapping failure probabilities must be kept strictly below the targeted IND-CPA-D security level. For instance, ensuring a failure probability below 2^{-128} is essential for achieving 128-bit IND-CPA-D security. We stress that this is not a practical break of HElib at its default parameters, where a failure probability of about 2^{-80} would demand on the order of 2^{80} bootstrappings; rather, our attack shows that these parameters offer only about 80 bits of IND-CPA-D security, well below the 128 targeted under IND-CPA. The experiments enlarge the failure probability solely to exhibit the mechanism within reach.

Regarding the second attack, we suggest three mitigations. First, noise-analysis frameworks should adopt worst-case error bounds rather than average-case behavior. Second, the a-component of a pt-ct product should be re-randomized by adding a fresh encryption of 0, with randomness derived by hashing the input ciphertext via SHA3, making the output distribution independent of the chosen plaintext while preserving homomorphic consistency. Third, one may simply encrypt the plaintext operand on the fly and fall back to a standard ct-ct multiplication. The tools behind the first two are not new: rigorous worst-case bounds and randomized rounding and lifting go back to the Ring-LWE toolkit of [LPR13]; our attacks exploit that they are not used in practice.

For the third attack, worst-case bounds again suffice but raise the evaluation cost; a lighter reactive measure is to increase the expansion factor δ_N by a margin absorbing the attack. Alternatively, one can reject and resample any secret key whose canonical embedding norm exceeds a fixed threshold, which blocks our use of ω.

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anonymous artifact reviewer for pointing out a discrepancy in the Hamming weight of the decryption key between Table 1 and the parameters used for Table 2, which has now been corrected.

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A Reminders

A.1 Bootstrapping Attack for DM/CGGI

We review the key-recovery attack on DM/CGGI from [CCP+24], as our BGV/BFV bootstrapping attack follows the same general design. In DM/CGGI, a ciphertext has the form $(b, a_0, \ldots, a_{n-1}) \in \mathbb{Z}_q^{n+1}$ with

$$b + \sum_{i \in [n]} a_i s_i = \Delta\mu + \varepsilon_{\text{in}} \pmod q \ ,$$

for some plaintext $\mu \in \mathbb{Z}_p$, scaling factor $\Delta = q/p$ (assuming $p \mid q$ for simplicity), and some noise term ε_{in} whose absolute value is small relative to Δ. Bootstrapping includes a modulus switching operation mapping $\text{ct} = (b, a_1, \ldots, a_n) \in \mathbb{Z}_q^{n+1}$ with modulus q to $\tilde{\text{ct}} = (\tilde{b}, \tilde{a}_1, \ldots, \tilde{a}_n) \in \mathbb{Z}_{2N}^{n+1}$ with modulus $2N$, where N is the

RLWE ring degree used in the rest of bootstrapping, such that

$$\tilde{b} = \left\lfloor \frac{2N \cdot b}{q} \right\rceil,\ \tilde{a}_i = \left\lfloor \frac{2N \cdot a_i}{q} \right\rceil,\ \text{for } i \in [n] \ .$$

Define the rounding errors as $\varepsilon_0 = \tilde{b} - 2Nb/q$ and $\varepsilon_i = \tilde{a}_i - 2Na_i/q$ for $i \in [n]$, and $\boldsymbol{\varepsilon} = (\varepsilon_0, \varepsilon_1, \ldots, \varepsilon_n) \in (-1/2, 1/2]^{n+1}$. Then, we have

$$\tilde{b} + \sum_{i \in [n]} \tilde{a}_i s_i = \frac{2N}{q}(\Delta\mu + \varepsilon_{\text{in}}) + \langle \boldsymbol{\varepsilon}, (1, \mathbf{s}) \rangle \pmod{2N} \ .$$

Here $\boldsymbol{\varepsilon}$ and the distribution of the secret key $\mathbf{s}$ are public, so assuming $\boldsymbol{\varepsilon} \sim \mathcal{U}([-1/2, 1/2)^{n+1})$ the distribution of the noise term $\langle \boldsymbol{\varepsilon}, (1, \mathbf{s}) \rangle$ is known. This term governs bootstrapping correctness: there exists $t > 0$ such that a failure implies

$$\langle \boldsymbol{\varepsilon}, (1, \mathbf{s}) \rangle + \frac{2N}{q} \cdot \varepsilon_{\text{in}} \geq t \ . \tag{12}$$

The attack of [CCP+24] exploits such failures: the value ε_i affects the failure probability if and only if $s_i \neq 0$. For $\mathbf{s} \in \{0, 1\}^n$, define $Y_i = \langle \boldsymbol{\varepsilon}, (1, \mathbf{s}) \rangle - \varepsilon_i s_i$ for $i \in [n]$, then the density function f_i of $\varepsilon_i \in [-1/2, 1/2)$ conditioned on (12) satisfies

$$f_i(x_i) = \begin{cases} \dfrac{\Pr[\varepsilon_i + Y_i + \frac{2N}{q}\varepsilon_{\text{in}} > t \mid \varepsilon_i = x_i]}{\Pr[\varepsilon_i + Y_i + \frac{2N}{q}\varepsilon_{\text{in}} > t]} & \text{if } s_i = 1 \ , \\ 1 & \text{if } s_i = 0 \ , \end{cases}$$

for $x_i \in [-1/2, 1/2]$. Assume that Y_i follows the Irwin-Hall distribution for $\mathrm{hw}(\mathbf{s})$ summations of independent copies of $\mathcal{U}([-1/2, 1/2])$, and that the collected failing ciphertexts are independent. Under these assumptions, one can numerically evaluate the conditional density f_i and decide whether $s_i = 0$ or $s_i = 1$, e.g., by estimating the closeness (with the KL divergence) between the observed distribution and the predicted conditional distributions.

A.2 Preliminaries for BGV and BFV

Let $[x]_q$ denote the reduction of x modulo q, and $\lfloor x \rceil$ denote the rounding of x to the nearest integer. Let $s \in \mathcal{R}$ be the secret polynomial.

A.2.1 The BFV scheme [FV12, KPZ21]. A BFV ciphertext $\mathrm{ct} = (a, b) \in \mathcal{R}_q^2$ encrypting a plaintext $\mu \in \mathcal{R}_t$ satisfies $b + as = \Delta\mu + \varepsilon \pmod{q}$, where $\Delta = \lfloor q/t \rfloor$ is the scaling factor and $\varepsilon \in \mathcal{R}$ is the noise. This can be written over $\mathcal{R}$ as $b + as = \Delta\mu + \varepsilon + qI$ for some quotient polynomial $I \in \mathcal{R}$.

Decryption. A decryption circuit $\mathrm{Dec}_s(\mathrm{ct})$ recovers its plaintext μ by computing

$$\mathrm{Dec}_s(\mathrm{ct}) = \left\lfloor \frac{t}{q} \cdot [b + as]_q \right\rceil \pmod{t} \ .$$

Decryption is correct if and only if $\|\varepsilon\|_\infty < q/(2t)$.

Additions. For two ciphertexts $\mathrm{ct}, \mathrm{ct}'$ encrypting μ, μ', the sum $([a + a']_q, [b + b']_q)$ encrypts $[\mu + \mu']_t$. The scaling factor does not distribute over the reduction modulo t: writing $\mu+\mu' = [\mu+\mu']_t + t \cdot u$ with $\|u\|_\infty \leq 1$, and $\Delta t = q - r_t(q)$ with $r_t(q) := q - t\lfloor q/t \rfloor$, one has $\Delta\mu + \Delta\mu' = \Delta[\mu + \mu']_t - r_t(q) \cdot u \pmod{q}$, so that the noise of the sum is $\varepsilon + \varepsilon' - r_t(q) \cdot u$ and

$$\|\varepsilon_{\text{add}}\|_\infty \leq \|\varepsilon\|_\infty + \|\varepsilon'\|_\infty + r_t(q) \ . \tag{13}$$

Iterating, a combination $\sum_i c_i \mathrm{ct}_i$ with integer coefficients has noise bounded by $\left(\sum_i |c_i|\right)V + \left(\sum_i |c_i| - 1\right) r_t(q)$ when all the $\|\varepsilon_i\|_\infty$ are at most V. The term $r_t(q)$, which is absent from BGV, is identified in [KPZ21, Section 3] as the source of the faster noise growth of BFV for large t.

Modulus Switching. For a ciphertext $\mathrm{ct} = (a, b) \in \mathcal{R}_q^2$, a modulus switching from the modulus q to a new modulus q' outputs

$$\mathrm{ct}' = (\lfloor q'a/q \rceil, \lfloor q'b/q \rceil) \in \mathcal{R}_{q'}^2 \ .$$

As utilized in bootstrapping, where $q' = p^e$ and $t = p^r$, the phase satisfies $b' + a's \approx (q'/t)\mu \pmod{q'}$, preserving the message in the most significant digits.

Multiplications. A pt-ct multiplication between μ' and $\mathrm{ct} = (a, b) \in \mathcal{R}_q^2$ is defined as

$$\mathsf{CMult}(\mu', \mathrm{ct}) = ([\mu' \cdot a]_q, [\mu' \cdot b]_q) \ .$$

A ciphertext-ciphertext multiplication, which includes relinearization, is defined as

$$\mathsf{Mult}_{\mathsf{rlk}}(\mathrm{ct}, \mathrm{ct}') = \mathsf{Relin}_{\mathsf{rlk}}\left(\left\lfloor \frac{t}{q} bb' \right\rceil_q, \left\lfloor \frac{t}{q}(ab' + a'b) \right\rceil_q, \left\lfloor \frac{t}{q} aa' \right\rceil_q \right) \ ,$$

where $\mathsf{Relin}_{\mathsf{rlk}} : \mathcal{R}_q^3 \to \mathcal{R}_q^2$ reduces the 3-element ciphertext after the tensor product to a 2-element ciphertext, using a relinearization key rlk. The noise $\varepsilon_{\mathsf{mul}}$ before and after relinearization is both dominated by

$$\mu\varepsilon' + \mu'\varepsilon + \frac{t}{q}\varepsilon\varepsilon' + t(\varepsilon I' + \varepsilon' I) \ ,$$

where (I, ε) and (I', ε') are the tuples of the rounding and noise terms, respectively for the ciphertexts (a, b) and (a', b').

Encoding in RNS Instantiations. In OpenFHE [Ope26], as an example of RNS implementation of BFV, the encoding do not scale by the constant $\lfloor q/t \rfloor$ only. They apply the scaling coefficient-wise, as

$$\Delta(\mu)_j = \mathrm{sign}(\mu_j) \left\lfloor \frac{q \cdot |\mu_j|}{t} \right\rceil \ .$$

Writing $\nu(\mu) := q\mu - t\Delta(\mu)$, one can get $\nu(\mu)_j = \mathrm{sign}(\mu_j)\left[q \cdot |\mu_j|\right]_t$, so that $\|\nu(\mu)\|_\infty < t$ and $\nu(0) = 0$.

Concretely, let $\mathrm{ct}_1, \ldots, \mathrm{ct}_M$ all encrypt the same μ, with noises $\varepsilon_1, \ldots, \varepsilon_M$ bounded by V, and let $c_1, \ldots, c_M$ be integers whose sum $C := \sum_i c_i = \rho t$ is a multiple of t. Then $t \cdot \Delta(\mu) = q\mu - \nu(\mu)$ gives

$$\sum_i c_i \cdot \Delta(\mu) = C \cdot \Delta(\mu) = \rho \cdot q \cdot \mu - \rho \cdot \nu(\mu) \equiv -\rho \cdot \nu(\mu) \bmod q \ ,$$

so that $\sum_i c_i \cdot \mathrm{ct}_i$ is an encryption of $[C \cdot \mu]_t = 0$, whatever μ, with noise

$$\sum_i c_i \varepsilon_i - \rho \cdot \nu(\mu) \ , \tag{14}$$

and $\|\rho \cdot \nu(\mu)\|_\infty < C$, so that the whole being bounded by $C \cdot (V + 1)$.

A.2.2 The BGV scheme [HS20, HS21]. Unlike BFV, the BGV scheme encodes the message in the least significant bits (LSB). A BGV ciphertext $\mathrm{ct} = (a, b) \in \mathcal{R}_q^2$ encrypting a plaintext $\mu \in \mathcal{R}_t$ satisfies $b + as = \varepsilon \pmod{q}$ and $\mu = [\varepsilon]_t$. Over $\mathcal{R}$, this is $b + as = \varepsilon + qI$ for some $I \in \mathcal{R}$.

Decryption. A decryption circuit $\mathrm{Dec}_s(\mathrm{ct})$ recovers its plaintext μ by computing

$$\mathrm{Dec}_s(\mathrm{ct}) = [[b + as]_q]_t \pmod{t} \ .$$

Decryption is correct if and only if $\|\varepsilon\|_\infty < q/2$.

Modulus Switching. For a ciphertext $\mathrm{ct} = (a, b) \in \mathcal{R}_q^2$, a modulus switching from the modulus q to a new modulus q' outputs

$$\mathrm{ct}' = \left(a' := \frac{q'a + t\delta_a}{q}, b' := \frac{q'b + t\delta_b}{q}\right) \in \mathcal{R}_{q'}^2 \ ,$$

preserving the message modulo t, where δ_i are correction terms. The resulting noise (or phase) $x = b' + a's$ can be written as $x = \varepsilon' + q'I$ over $\mathcal{R}$ for the same $I \in \mathcal{R}$ for ct, where

$$\varepsilon' = \frac{q'}{q}\varepsilon + t(\varepsilon_b + \varepsilon_a s) \ ,$$

with $\varepsilon_a = \delta_a/q$ and $\varepsilon_b = \delta_b/q$. This ensures the plaintext is preserved as $[\varepsilon']_t = [(q'/q)\mu]_t \pmod{t}$.

Multiplications. A pt-ct multiplication between μ' and $\mathrm{ct} = (a, b) \in \mathcal{R}_q^2$ is defined as

$$\left([\mu' \cdot a]_q, [\mu' \cdot b]_q\right) \ .$$

The ciphertext-ciphertext multiplication $\mathsf{Mult}_{\mathsf{rlk}}(\mathrm{ct}, \mathrm{ct}')$ first computes the tensor product $(d_0, d_1, d_2) = (bb', ab' + a'b, aa') \in \mathcal{R}_q^3$. This intermediate output satisfies $d_0 + d_1 s + d_2 s^2 = \varepsilon_{\mathsf{mul}} \pmod{q}$, where the noise $\varepsilon_{\mathsf{mul}}$ is given by:

$$\varepsilon_{\mathsf{mul}} = (\varepsilon + qI)(\varepsilon' + qI') = \varepsilon\varepsilon' + q(\varepsilon I' + \varepsilon' I + qII') \ .$$

Note that $\varepsilon_{\mathsf{mul}} \equiv \varepsilon\varepsilon' \pmod{q}$. The relinearization procedure $\mathsf{Relin}_{\mathsf{rlk}}$ is then applied to reduce the result back to two elements while preserving the noise relation modulo q. Unlike BFV, BGV multiplication causes the noise to grow quadratically as $\varepsilon_{\mathsf{mul}} \approx \varepsilon \cdot \varepsilon'$. To manage this growth, the multiplications are typically followed by the modulus switching step. By switching from q to a smaller modulus q', the noise $\varepsilon_{\mathsf{mul}}$ is scaled down to $\varepsilon' \approx (q'/q)\varepsilon_{\mathsf{mul}}$, effectively keeping the noise-to-modulus ratio stable.

A.3 Probabilistic Bounds

A.3.1 Real Gaussian tail. For a real Gaussian random variable $X \sim \mathcal{N}(0, \sigma^2)$ and any $k > 0$,

$$\Pr[\,|X| > k\sigma\,] = \mathrm{erfc}\left(k/\sqrt{2}\right), \tag{15}$$

where erfc denotes the complementary error function

$$\mathrm{erfc}(z) := 1 - \frac{2}{\sqrt{\pi}}\int_0^z \exp(-t^2)\,dt = \frac{2}{\sqrt{\pi}}\int_z^\infty \exp(-t^2)\,dt.$$

A.3.2 Evaluation at primitive roots of unity. Let $\mathcal{R} = \mathbb{Z}[X]/(X^N + 1)$ for a power-of-two N, and let $\omega \in \mathbb{C}$ be a primitive $(2N)$-th root of unity. Consider $a = \sum_{i\in[N]} a_i X^i$ whose coefficients $\{a_i\}$ are i.i.d variables with zero mean and variance σ^2. By the Central Limit Theorem, $a(\omega) = \sum_i a_i \cdot \omega^i$ is heuristically modeled as a complex Gaussian distribution $\mathcal{CN}(0, N\sigma^2)$ for sufficiently large N, where the circular symmetry follows from the even distribution of the $(2N)$-th roots of unity on the unit circle. We refer to [HS20, Section 2.6.1] for more details.

Consequently, $|a(\omega)|$ follows a Rayleigh distribution with scale parameter $\sqrt{N/2} \cdot \sigma$, and a standard polar-coordinate computation yields

$$\Pr\Big[\ |a(\omega)| > k\sqrt{N} \cdot \sigma\ \Big] = \exp(-k^2) \quad \text{for any } k > 0. \tag{16}$$

For instances, if $a \leftarrow \mathcal{U}([-1/2, 1/2))^N$, each coefficient has variance $\sigma^2 = 1/12$, so

$$\Pr\left[\ |a(\omega)| > k\sqrt{\frac{N}{12}}\ \right] = \exp(-k^2).$$

For $a \leftarrow \mathcal{U}(\{-1, 0, 1\})^N$, each coefficient has variance $\sigma^2 = 2/3$, so

$$\Pr\left[\ |a(\omega)| > k\sqrt{\frac{2N}{3}}\ \right] = \exp(-k^2).$$

A.3.3 Canonical norm. The canonical norm $\|a\|_\infty^{\mathsf{can}} = \max_i |a(\omega_i)|$ is the maximum of $|a(\omega_i)|$ over the $N/2$ conjugate pairs of primitive $2N$th roots of unity (since a has real coefficients, $a(\bar{\omega}) = \overline{a(\omega)}$). Applying the union bound over these $N/2$ pairs,

$$\Pr\Big[\ \|a\|_\infty^{\mathsf{can}} > k\sqrt{N} \cdot \sigma\ \Big] \le \frac{N}{2} \cdot \exp(-k^2). \tag{17}$$

Heuristically modeling the $N/2$ values as independent yields the sharper estimate

$$\Pr\Big[\ \|a\|_\infty^{\mathsf{can}} \le k\sqrt{N} \cdot \sigma\ \Big] \approx \left(1 - \exp(-k^2)\right)^{N/2},$$

which concentrates around $k \approx \sqrt{\ln(N/2)}$, i.e., $\|a\|_\infty^{\mathsf{can}}$ is typically of order $\sqrt{N\ln(N/2)} \cdot \sigma$ for large N.

B Choice of Plaintext for Each Primitive Root of Unity

In this section, we show how to build plaintexts with large evaluations in the primitive $(2N)$-th root of unity, as required in Section 4.1.

For $k \in [N]$, let $\omega_k = \exp\left((2k-1)\pi\mathrm{i}/N\right)$. We now construct a plaintext μ with $\|\mu\|_\infty = 1$ while $|\mu(\omega_k)| \approx 2N/\pi$. Define $\mu = \sum_{0\le j<N} \mu_j X^j$ with

$$\mu_j = \mathrm{sign}\left(\cos\left(\pi\frac{(2k-1)\ (2j+1)}{2N}\right)\right) \ . \tag{18}$$

Then we have

$$\mu(\omega_k) = \sum_{0\le j<N} \mathrm{sign}\left(\cos\left(\pi\frac{(2k-1)\,(2j+1)}{2N}\right)\right) \cdot \exp\left(\mathrm{i}\pi\frac{(2k-1)j}{N}\right),$$

where $\mathrm{i} = \sqrt{-1}$. Since $2k-1$ is co-prime to $2N$, the set of angles

$$\left\{\pi\frac{(2k-1)(2j+1)}{2N} \pmod{\pi} : j = 0, \ldots, N-1\right\}$$

is exactly the set of odd multiples of $\pi/(2N)$ in $[0, \pi)$; this suffices, as each summand is unchanged when its angle is shifted by π. Due to the symmetries of the cosine and sine functions, the imaginary part of $\mu(\omega_k) \cdot e^{\mathrm{i}\pi(2k-1)/(2N)}$ vanishes exactly, and the magnitude is given by:

$$|\mu(\omega_k)| = \left|\mu(\omega_k) \cdot e^{\mathrm{i}\pi\frac{(2k-1)}{2N}}\right| = \sum_{0\le j<N}\left|\cos\left(\pi\frac{(2j+1)}{2N}\right)\right| \ .$$

We have

$$\sum_{0\le j<N}\left|\cos\left(\pi\frac{(2j+1)}{2N}\right)\right| = 2\sum_{0\le j<N/2}\cos\left(\pi\frac{(2j+1)}{2N}\right) \ ,$$

and by multiplying $\sin(\pi/(2N))$, we obtain

$$\sin\left(\frac{\pi}{2N}\right)\cdot|\mu(\omega_k)| = 2\sum_{0\le j<N/2}\sin\left(\frac{\pi}{2N}\right)\cos\left(\pi\frac{(2j+1)}{2N}\right)$$
$$= \sum_{0\le j<N/2}\left[\sin\left(\pi\frac{(j+1)}{N}\right)-\sin\left(\pi\frac{j}{N}\right)\right] .$$

Recognizing a telescoping series, this then simplifies to:

$$\sin\left(\pi\frac{(N/2)}{N}\right)-\sin(0)=\sin\left(\frac{\pi}{2}\right)=1 .$$

Thus, we obtain the exact magnitude:

$$|\mu(\omega_k)| = \frac{1}{\sin\left(\frac{\pi}{2N}\right)} ,$$

which can be approximated to $2N/\pi$ when N grows.

In the experiments with the OpenFHE library, where negative coefficients are not allowed for μ, we considered $\mu' = \sum_{0\le j<N}\mu'_j X^j$, whose j-th coefficient μ'_j is 0 if $\mu_j < 0$, and otherwise μ_j. We used μ' for every $k \in [N]$. As $\mu'_j = (1+\mu_j)/2$, we have:

$$\mu' = \frac{1}{2}\left(\mu + \sum_{0\le j<N} X^j\right) .$$

Now, a computation similar to the one performed for $e^{i\pi/N}$ in Section 4.1 gives that

$$\left|\sum_{0\le j<N}\omega_k^j\right| = \frac{1}{\left|\sin\left(\pi\frac{2k-1}{2N}\right)\right|} .$$

By the triangle inequality, we thus obtain that

$$|\mu'(\omega_k)| \ge \frac{1}{2}\left(|\mu(\omega_k)| - \left|\sum_{0\le j<N}\omega_k^j\right|\right)$$
$$= \frac{1}{2}\left(\frac{1}{\sin(\frac{\pi}{2N})} - \frac{1}{\left|\sin\left(\pi\frac{2k-1}{2N}\right)\right|}\right) .$$

The lower bound is minimized for $k = 2$ and $k = N-1$, and, in that case, it is $\approx 2N/(3\pi)$. It is vacuous for $k = 1$ and $k = N$, where μ' is $\sum_{0\le j<N/2} X^j$ (for $k = 1$) or $X^{N/2}$ times it (for $k = N$), and a direct computation gives $|\mu'(\omega_k)| = |1-\omega_k^{N/2}|/|1-\omega_k| = 1/(\sqrt{2}\sin(\frac{\pi}{2N})) \approx \sqrt{2}N/\pi$.

C More Bootstrapping Experiments

As in Figure 2 in Section 3, we present the distribution of the coefficients c_j conditioned on failure for an additional parameter set B* in Figure 6. We collected 354 ciphertext failures, comprising a total of 17,582 failed slots. For each graph, we used over 47,000 data points, except for the one in subfigure (a), which is based on 17,582 points. As in Figure 2, the graphs show a clear distinction between $s_j = 0$ and $s_j = \pm 1$.

In addition, we present the distribution of the variances $\mathrm{Var}(c_j)$ we computed from the obtained failing slots, for each $\ell = 0, 1, 2, 3$ and for each set of parameters A* and B* in Figures 7 and 8, respectively. When the number of failed slots is sufficiently large, the unusually large variances serve as strong indicators of the corresponding non-zero indices.

D More Error Growth Experiments

In Figure 9, we present another instance of the BFV pt-ct multiplication attack from Section 4.2, focusing on the noise growth. In Figure 10, we present the full trends of the noise growth during the BFV ct-ct multiplication attack, omitted from Figure 4.

E More CT-CT Attack Results

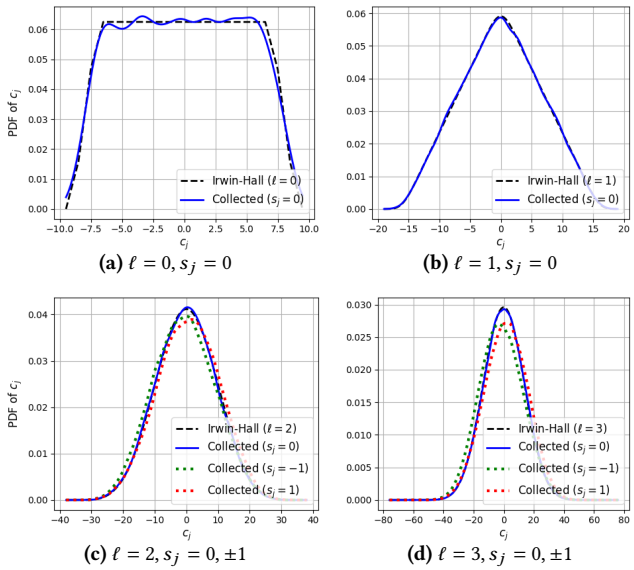
E.1 Noise Behavior

Table 4 summarizes the noise and the decryption result obtained from 40 runs for each $\mathcal{D}_0$ and $\mathcal{D}_1$.

Table 4: Noise before and after the K squarings, averaged over the same 40 runs. The last column shows the maximum observed decrypted result y_j over the indices j, per run. The first three columns are in $\log_2$-scale, and brackets give the observed range.

	Noise, before		*Noise, after*	*Decrypted*
	$\|\varepsilon_0\|_\infty$	$\|\varepsilon_0(\omega)\|$	$\|\varepsilon_K\|_\infty$	$\max_j \|y_j\|$
$\mathcal{D}_0$	37.21	43.93	1,334.4 [1,332.2, 1,336.0]	4 [1, 11]
$\mathcal{D}_1$	37.29	48.51	1,338.9 [1,338.8, 1,339.0]	90 [84, 97]

Before the amplification, the two are a factor ≈ 1.05 apart in the infinity norm, while their phases at ω_{succ} already differ by a factor ≈ 24; the squaring iteration turns the latter into the factor $90/4 \approx 23$ observed in the decryption results. All $80 = 40 + 40$ runs end in a decryption failure, for a fixed key pair. Note that, both distributions are centered at 0.



(a) $\ell = 0, s_j = 0$ **(b)** $\ell = 1, s_j = 0$

(c) $\ell = 2, s_j = 0, \pm 1$ **(d)** $\ell = 3, s_j = 0, \pm 1$

Figure 6: Distributions of the coefficients c_j at index j, conditioned on bootstrapping failure, for $\ell \in \{0, 1, 2, 3\}$ and $s_j \in \{-1, 0, 1\}$. Dashed black curves show the PDFs of $\mathcal{D}^{(\ell)}$, while solid blue, dotted green, and dotted red curves show the empirical PDFs of the collected coefficients c_j from failing ciphertexts, where $s_j = 0, -1$, and 1, respectively (parameter set B*).

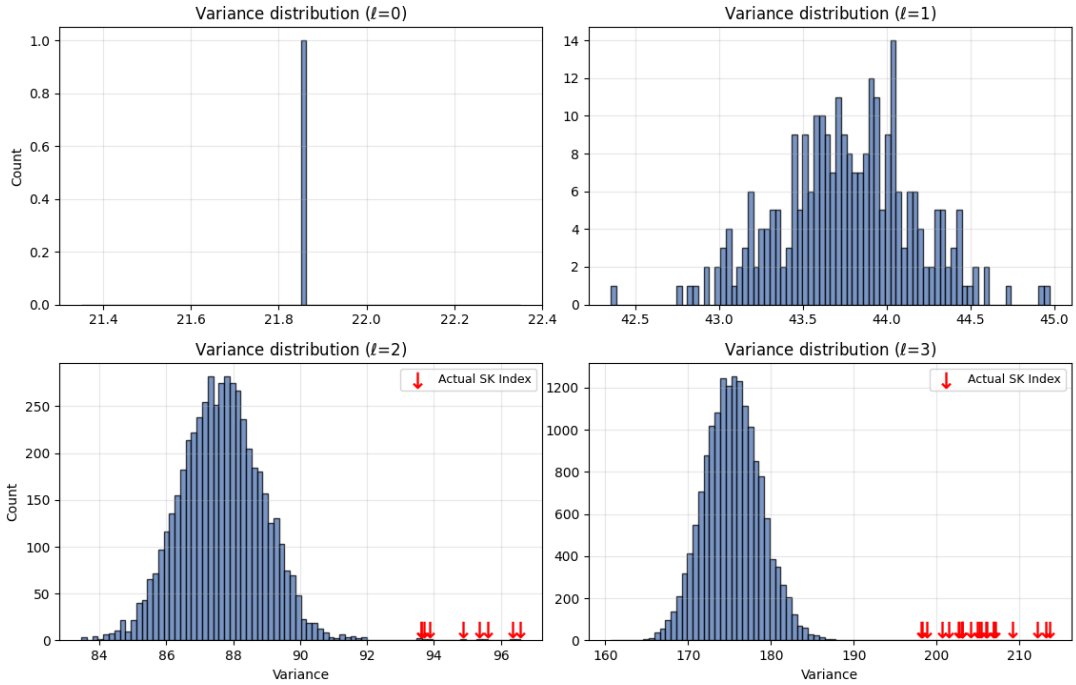


Figure 7: Distribution of heuristic variances $\mathrm{Var}(c_j)$ for each $\ell = 0, 1, 2, 3$, computed from the 23,800 failing slots, collected from parameter set A*. The red arrows indicates the variances corresponding to the non-zero secret key indices.

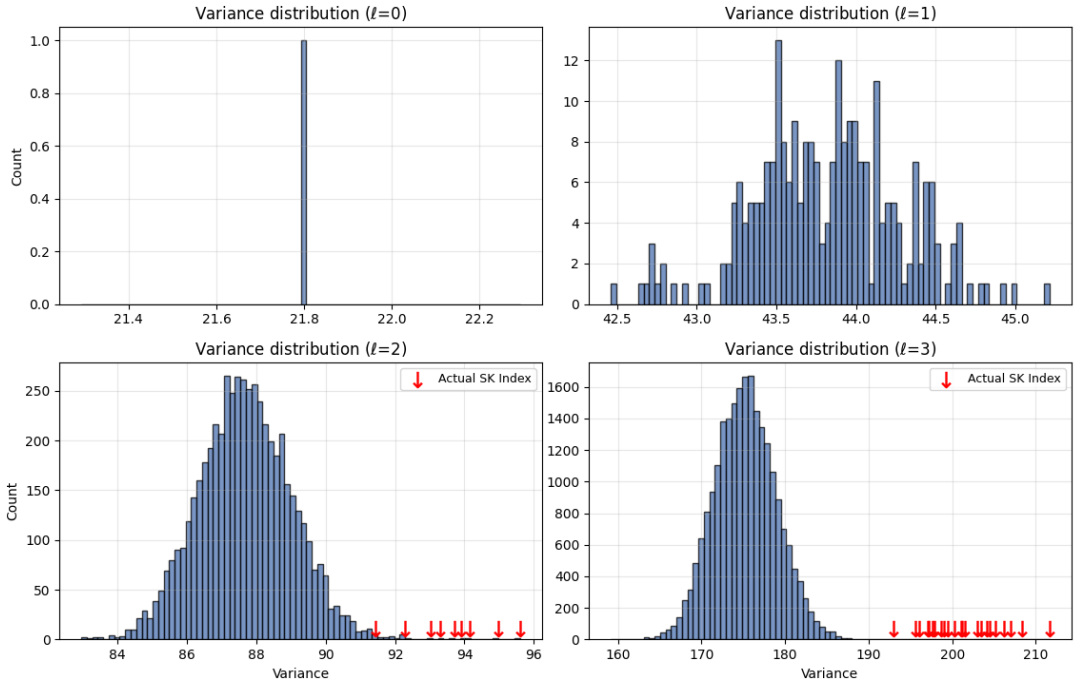


Figure 8: Distribution of heuristic variances $\mathrm{Var}(c_j)$ for each $\ell = 0, 1, 2, 3$, computed from the 17,582 failing slots, collected from parameter set B*. The red arrows indicates the variances corresponding to the non-zero secret key indices.

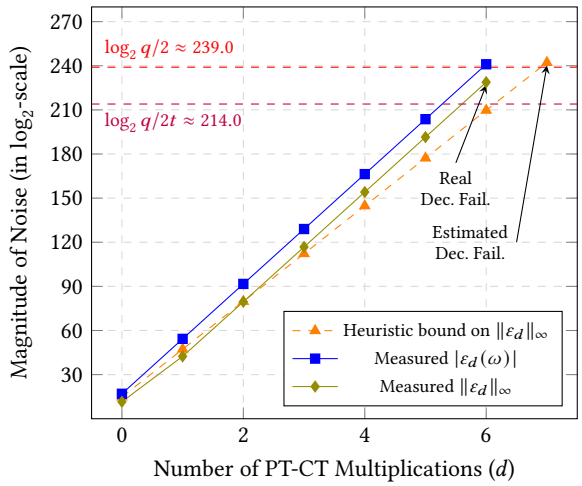


Figure 9: Noise Growth in Algorithm 1. $N = 2^{15}$, $q \approx 2^{240}$, $t \approx 2^{25}$, **index** $k = 7$, $\|\mu\|_\infty \approx 2^{24.0}$, **and** $|\mu(\omega)| \approx 2^{37.3}$.

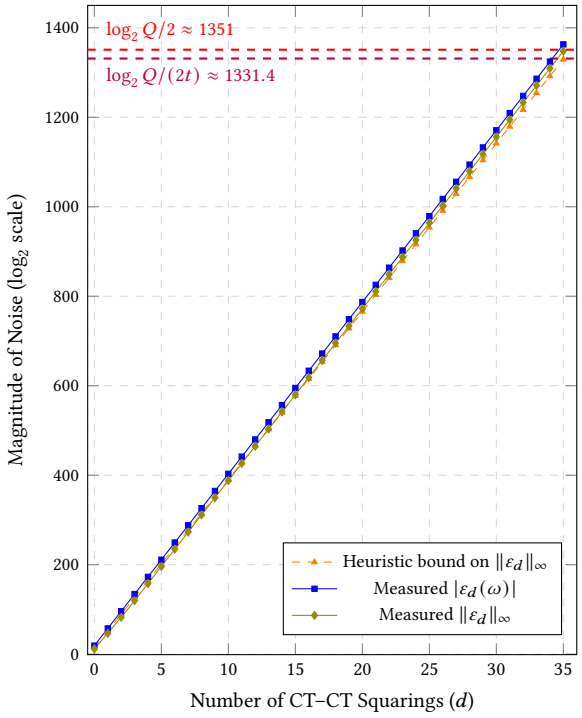


Figure 10: Full Noise Growth Trends in BFV CT-CT Squarings ($d = 0, \ldots, 35$). Parameters are $N = 2^{16}$, $\log_2 Q = 1{,}352$, and $\log_2 t = 19.585$. The measured noise $|\varepsilon_d(\omega)|$ outpaces the heuristic bound and eventually exceeds the decryption threshold at $d = 35$.

E.2 Independent Key Pairs

The key error source, $\rho \cdot v(\mathcal{H})$, does not depend on the key pairs, whereas the amplification factor $A = \prod_{d \le K} 2t \cdot (a_d^*/q)(\omega) \cdot s(\omega)$ does, through $s(\omega)$. We therefore repeated the attack over 10 independently generated key pairs, so that sk, pk, and every encryption randomness would differ, with three runs for each pair. We summarize the result in Table 5.

Table 5: Ten independently generated key pairs, three runs each. y_j denotes the observed decrypted result at index j.

k_s	$\max_j \lvert y_j \rvert$		$\log_2 \lvert \varepsilon_0(\omega) \rvert$	
	$\mathcal{D}_0$	$\mathcal{D}_1$	$\mathcal{D}_0$	$\mathcal{D}_1$
3.382	19	212	44.01	48.57
3.492	56	624	44.77	48.57
3.281	4	74	43.90	48.53
3.403	30	258	44.23	48.53
3.421	28	299	43.35	48.49
3.191	2	28	43.58	48.51
3.272	6	68	44.48	48.54
3.413	5	290	41.73	48.50
3.689	233	4,189	43.79	48.61
3.246	3	53	44.13	48.55

All ten experiments show a clear separation in $\mathcal{D}_0$ and $\mathcal{D}_1$, by a factor between 8.6 and 58. We note that, depending on k_s, the two distributions also scale, which in turn requires the reference distributions of Phase 2 to distinguish the challenge distribution in the end.

E.3 Cost in Revised OpenFHE

The revised heuristic bound $(8Nt)^K \cdot V_N$ replaces δ_N^2 by $\delta_N \cdot \delta_{\mathrm{ms},N}$, so that the admissibility condition of Inequality (9) becomes $k_a k_s > 12\sqrt{2}$. This per-step condition is however not sufficient by itself, as the gain $k_a k_s / 12\sqrt{2}$ must also, over the K steps, recover the handicap the adversary starts with.

The adversary does not know the exact k_s, so we set a highly probable window for k_s; then set C and k_a to work when the secret sk lies in that window.

Table 6: Cost of the ct-ct attack against the revised bound, for $N = 2^{16}$. Encryption queries are counted as $\log_2$ of the number of trials, since a pool of L encryptions of 0 yields 2^L subset sums, plus the $M = 2^{18}$ encryptions consumed by Phases 2 and 3. Decryption queries cover the $N/2$ candidates for ω. Trials involve no oracle access, and count the K re-scans of the pool, that is $K \exp(k_a^2)$.

t	K	k_s	k_a	#trials	#enc.	#dec.
1,153	44	3.28	8.09	$2^{99.9}$	2^{18}	2^{15}
		3.40	7.81	$2^{93.5}$	2^{18}	2^{15}
		3.50	7.58	$2^{88.4}$	2^{18}	2^{15}
257	47	3.28	7.87	$2^{94.9}$	2^{18}	2^{15}
		3.40	7.59	$2^{88.7}$	2^{18}	2^{15}
		3.50	7.37	$2^{83.9}$	2^{18}	2^{15}

Table 6 shows the three cost factors for two candidate parameter sets, both at $N = 2^{16}$ and both claiming 128 bits of IND-CPA security.

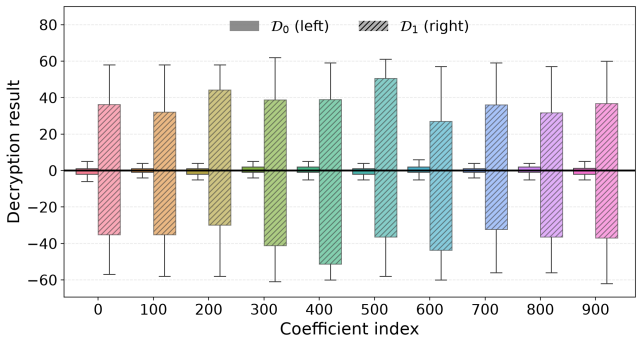


Figure 11: The same experiment as Figure 5, against the revised bound of the current OpenFHE: $\mathcal{D}_0$ (left of each pair, solid) and $\mathcal{D}_1$ (right, hatched) over 40 runs, at $N = 2^{16}$, $t = 1{,}153$, $k_a = 8.15$, $k_s = 3.425$, $M = 2^{18}$, $C = 2^{33.23}$ and $K = 44$. The certified bound is $2^{1331.03}$ against $q/2t = 2^{1340.83}$, so the decryption oracle answers; the noise actually reached is $2^{1342.97}$ and $2^{1347.66}$, above that threshold and below $q/2 = 2^{1351}$.

As the adversary succeeds only when the key meets the target, its advantage is the probability p of that event, halved; dividing the trials by it turns the second row of each block into $2^{96.4}$ and $2^{91.6}$, still far below the 2^{128} claimed. Balancing $k_a \approx k_s$ lowers this further, to $2^{68.0}$ at $k_a = k_s = 5.15$ for $t = 1{,}153$; we do not put that forward, as it is easily defeated by rejecting atypical secret keys at key generation.

We mounted the attack against OpenFHE v1.5.0 for both parameter sets. For $t = 1{,}153$ we take $k_a = 8.15$ and draw the key until k_s falls in $[3.40, 3.48]$, a band a random key meets with probability 0.10; the reciprocal of that probability multiplies the cost, and the key we kept realizes $k_s = 3.425$. Selecting the key is, of course, not something an adversary can do, and Table 7 below dispenses with it. Taking $M = 2^{18}$ and $C = \rho t = 2^{33.23}$, the certified bound is $2^{1331.03}$, below $q/2t = 2^{1340.83}$: the decryption oracle answers. The noise actually reached averages $2^{1342.97}$ for $\mathcal{D}_0$ and $2^{1347.66}$ for $\mathcal{D}_1$, in both cases above the threshold and below $q/2 = 2^{1351}$ so that no wrap-around occurs. Over 40 runs, $\max_j |y_j|$ lies in an interval $[0, 6]$ for $\mathcal{D}_0$ and in an interval $[50, 62]$ for $\mathcal{D}_1$; the two are separated without overlap. Figure 11 draws the two distributions, which behave exactly as those of Figure 5: both are centered at 0 and are distinguished by their spread, here by a factor 27.4 in inter-quartile range.

For $t = 257$, the key satisfied $k_s = 3.527$ and we take $C = 2^{45.47}$, leaving a margin of 14 bits below $q/2t = 2^{1342.99}$. 10 runs then give $\max_j |y_j| \in [0, 2]$ for $\mathcal{D}_0$ against $[14, 16]$ for $\mathcal{D}_1$, again without overlap.

Table 7 repeats the experiment over 10 independently generated key pairs, 10 runs for each key, with $k_a = 8.15$ and C fixed in advance.

Our C is set so that the attack succeeds if $3.08 < k_s < 3.89$: for $k_s < 3.08$, neither distribution reaches the threshold $q/2t$, so the decryption gives no information; and for $k_s > 3.89$, the noise exceeds $q/2$ and wraps around in both cases, so the decryption gives uniform-like coefficients over $[-t/2, t/2]$. A random secret key lies in that band with probability ≈ 0.91. 8 keys among the 10 keys fall inside it and separate without overlap, with the magnitude of $\mathcal{D}_1$ growing with k_s as expected; the two that do not are the two drawn below 3.08, which is what the band predicts.

Table 7: Ten independently generated key pairs against the revised bound, ten runs each, at $t = 1{,}153$ and $K = 44$ with $k_a = 8.15$ and $C = 2^{33.23}$ fixed beforehand. The third is the largest $\max_j |y_j|$ over the 10 runs of $\mathcal{D}_0$, and the fourth column is the smallest $\max_j |y_j|$ over the 10 runs of $\mathcal{D}_1$. The next two columns give the root mean square (rms) of the decrypted coefficients, averaged over the runs. Rows are sorted by k_s.

k_s	$\max_j \lvert y_j \rvert$		rms		outcome
	$\mathcal{D}_0$ (max)	$\mathcal{D}_1$ (min)	$\mathcal{D}_0$	$\mathcal{D}_1$	
3.039	0	0	0.0	0.0	undershoot
3.051	0	0	0.0	0.0	undershoot
3.135	0	1	0.0	0.9	separated, $\mathcal{D}_0$ not failed
3.212	0	3	0.0	2.3	separated, $\mathcal{D}_0$ not failed
3.256	0	6	0.0	4.4	separated, $\mathcal{D}_0$ not failed
3.325	1	15	0.5	11.1	separated by magnitude
3.376	4	27	1.3	21.0	separated by magnitude
3.420	6	52	2.4	38.7	separated by magnitude
3.581	40	377	18.2	291.9	separated by magnitude
3.605	50	517	21.1	394.1	separated by magnitude

F Bootstrapping Attack with $(h, \tilde{h}) = (120, 32)$

To evaluate the effect of the decryption key weight h, we repeated the experiments of Table 2 using parameters $A^{\#}$ and $B^{\#}$, which take $(h, \tilde{h}) = (120, 32)$. These are identical to A^* and B^* except that the decryption key weight is set to the library's default $h = 120$, while keeping $\tilde{h} = 32$ for high failure probability. Since the attack essentially depends only on the bootstrapping secret key, h plays almost no role: the per-slot failure probabilities depend solely on $\tilde{h}$ (remaining identical to those in Table 2), and the estimated attack cost remains unchanged.

We conducted four independent runs for each of $A^{\#}$ and $B^{\#}$, using a freshly generated key pair and collecting 500 failing ciphertexts per run. Table 8 summarizes the results as a function of the number of failing ciphertexts used. For each parameter set, the table reports the average number of failed slots across the four runs and, for each individual run, the number of the $\tilde{h} = 32$ nonzero indices recovered within the $3\tilde{h} + 1 = 97$ candidate set. In every run, all 32 nonzero indices are included once the number of failing ciphertexts reaches 400 for $A^{\#}$ and 450 for $B^{\#}$.

G Open Science

Our code is available at https://github.com/hmchoe0528/NEW_INDCPAD_HE. It includes the implementations for the following attacks and experiments. For more details, we refer to the documentation file `README.md` in the repository.

- BGV Bootstrapping KR-D Attack using HElib (Section 3). The attack collects secret keys, failing coefficients, and Irwin-Hall mapping for each index. The attack then recovers the secret key from the collected data. The plots in Figures 2, 6, 7, and 8 can also be generated using the collected data.

Table 8: Bootstrapping attack with $(h, \tilde{h}) = (120, 32)$, four runs with independently drawn keys for each parameter set. The rows show the number of failed slots, averaged over the four runs, and for each run the number of the $\tilde{h} = 32$ nonzero indices included in the $3\tilde{h} + 1 = 97$ candidates; shaded cells mark that all of them are.

	#ctxt used		50	100	150	200	250	300	350	400	450	500
$A^{\#}$	#slots used		2,619	5,290	7,921	10,649	13,215	16,041	18,744	21,323	24,217	26,949
	#correct	run 1	12	18	26	28	30	31	31	32	32	32
		run 2	8	19	29	32	32	32	32	32	32	32
		run 3	12	21	23	23	30	32	31	32	32	32
		run 4	7	15	24	28	31	31	32	32	32	32
$B^{\#}$	#slots used		3,001	5,846	8,688	11,629	14,502	17,265	20,321	23,294	26,078	28,876
	#correct	run 1	10	18	22	26	31	30	30	31	32	32
		run 2	7	15	21	26	30	31	31	32	32	32
		run 3	3	14	25	28	32	32	32	32	32	32
		run 4	4	11	22	28	29	32	32	32	32	32

- BFV PT-CT Multiplication KR-D Attack using OpenFHE (Section 4). The attack collects the decryption results. The secret key is then fully recovered. The scripts also generate raw data for Figures 3 and 9.
- BFV CT-CT Multiplication IND-CPA-D Attack using OpenFHE (Section 5). The attack runs the IND-CPA-D attack until phase 2, resulting in a distinguishable gap in the error distribution. The scripts also generate raw data for Figures 4, 5, and 10.

H Generative AI Usage

The authors used Gemini and Grammarly for language editing, and GitHub Copilot, Claude, and Gemini to assist the implementations; for the attack of Section 5, Claude identified an error in an earlier version and contributed to the fix. All AI outputs were verified by the authors, who take full responsibility for the paper.

I Ethical Considerations

This work analyzes the IND-CPA-D security of BGV and BFV schemes. While current FHE deployments primarily rely on the IND-CPA model, under which our findings do not compromise existing security, the transition toward multi-party settings, such as Threshold FHE, and recent NIST standardization efforts make IND-CPA-D security a critical concern for the community. We identify vulnerabilities not in implementation bugs, but in the widely accepted "heuristic noise analysis" paradigms commonly used for parameter selection.

To minimize potential harm, we provide comprehensive mitigations in the Conclusion and are committed to a coordinated vulnerability disclosure (CVD) process, engaging relevant stakeholders to enable protections prior to broad dissemination.